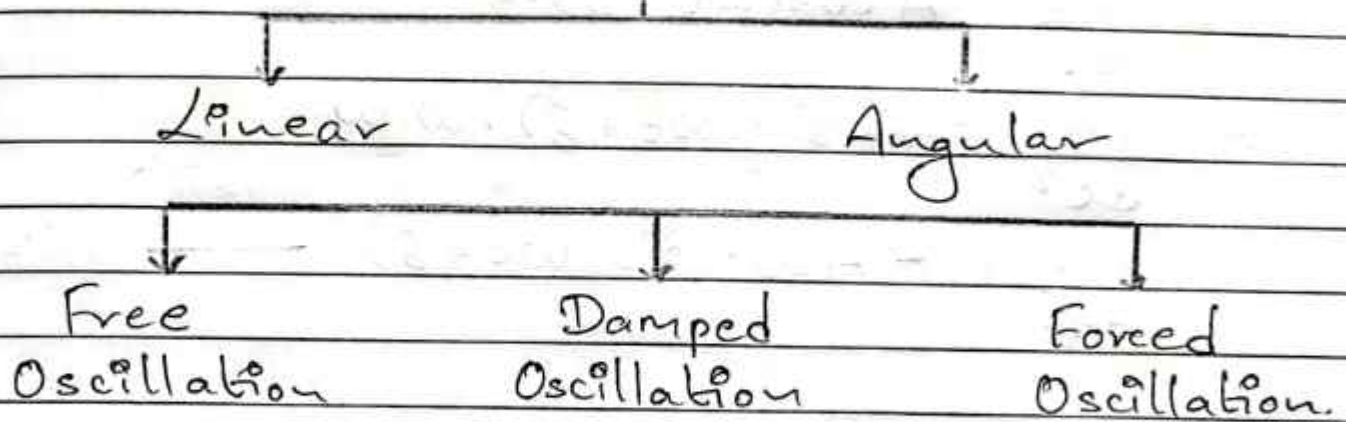


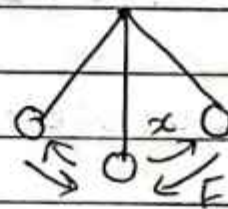
Unit - 1

Mechanical Oscillation



1) Free Oscillation

$$F \propto x$$
$$\text{or } F = -kx \quad \text{--- (1)}$$



$$\text{or, } ma = -kx$$

$$\text{or, } a = -\frac{k}{m} x \quad \text{--- (2)}$$

$$\frac{d^2x}{dt^2} = -\frac{k}{m} x$$

$$\frac{d^2x}{dt^2} + \frac{k}{m} x = 0 \quad \text{--- (3)}$$

Solⁿ of eqⁿ (3),

$$x(t) = A \sin(\omega t + \phi) \quad \text{--- (4)}$$

Diff. w.r.t. time 't',

$$x(t) = A \sin(\omega t + \delta)$$

$$\frac{dx}{dt} = A \cos(\omega t + \delta) \cdot \omega$$

$$\text{or } \frac{d^2x}{dt^2} = A \sin(\omega t + \delta) \cdot \omega \cdot \omega$$

$$\therefore \frac{d^2x}{dt^2} = -A\omega^2 \sin(\omega t + \delta) \quad \text{---}$$

Now, eqⁿ (3) becomes

$$-A\omega^2 \sin(\omega t + \delta) + \frac{k}{m} x = 0$$

$$\text{or } -A\omega^2 \sin(\omega t + \delta) + \frac{k}{m} A \sin(\omega t + \delta) = 0$$

$$\text{or } \frac{k}{m} A \sin(\omega t + \delta) = A\omega^2 \sin(\omega t + \delta)$$

$$\text{or } \frac{k}{m} = \omega^2 \quad \text{--- (5)}$$

$$\therefore \omega = \sqrt{\frac{k}{m}} \quad \text{--- (6)}$$

Terms associated with free Oscillation.

1) Amplitude, (A)

$$x = A \sin(\omega t + \delta) \quad \text{--- (1)}$$

$$x_{\max} = ?$$

Now,

$$\text{We know, } \sin(\omega t + \delta)_{\max} = \pm 1$$

$$\therefore x_{\max} = A \cdot (\pm 1) = \pm A$$

2) Time period

→ If 'T' be the time period then

$$x = A \sin(\omega t + \delta)$$

or,

$$x = A \sin[\omega(T + t) + \delta] \quad \text{--- (2)}$$

or,

$$x = A \sin[\omega t + \delta + 2\pi] \quad \text{--- (3)}$$

$$[\omega T = 2\pi f \cdot \frac{1}{f} = 2\pi]$$

Equating (2) & (3),

$$\omega(t + T) + \delta = \omega t + \delta + 2\pi$$

$$\omega T = 2\pi$$

$$\therefore T = \frac{2\pi}{\omega}$$

$$\text{--- (4)}$$

3) Frequency

$$\begin{aligned} T &\longrightarrow 1 \text{ sec} \\ 1 \text{ sec} &\longrightarrow \frac{1}{T} \end{aligned}$$

$$f = \frac{1}{T}$$

$$\therefore f = \frac{\omega}{2\pi} \quad (5)$$

Again,

$$\omega = 2\pi f$$

$$\omega = \frac{2\pi}{T} = \text{angular velocity} \\ \text{or} \\ \text{angular frequency}$$

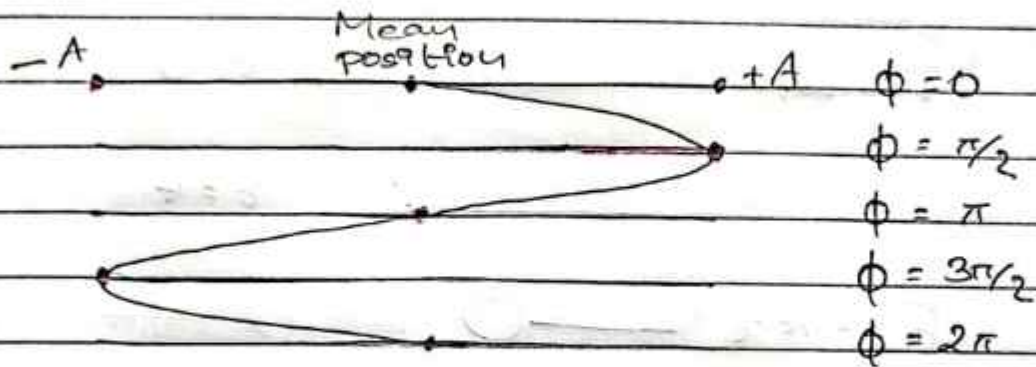
4) Phase & Phase Constant

$$x = A \sin(\omega t + \delta)$$

$$\text{phase}(\phi) = \omega t + \delta$$

$$t=0, \phi=0$$

Graphical representation of Phase,



Mathematical Analysis of Free Oscillation

Mean position x_0, v_0 $t=0$

x, v $t=t$

$$F = -kx \quad \text{--- (1)}$$

$$\text{or, } ma = -kx$$

$$\text{or, } m \cdot \frac{dv}{dt} = -kx$$

$$\text{or, } \frac{dv}{dt} = -\frac{k}{m} x \quad \text{--- (2)}$$

$$\text{or, } \frac{k}{m} = \omega^2 \quad \text{--- (3)}$$

$$\therefore \frac{dv}{dt} = -\omega^2 x \quad \text{--- (4)}$$

$$\frac{dv}{dx} \cdot \frac{dx}{dt} = -\omega^2 x$$

$$\int_{v_0}^v v \cdot dv = \int_{x_0}^x -\omega^2 x \cdot dx \quad \text{--- (5)}$$

$$\text{or } \left[\frac{v^2}{2} \right]_{v_0}^v = -\omega^2 \left[\frac{x^2}{2} \right]_{x_0}^x$$

$$\text{or } v^2 - v_0^2 = -\omega^2 x^2 + \omega^2 x_0^2$$

$$\text{or } v^2 = v_0^2 + \omega^2 x_0^2 - \omega^2 x^2$$

$$\text{or, } v = \omega \sqrt{\frac{v_0^2}{\omega^2} + x_0^2 - x^2} \quad \text{--- (6)}$$

or, $v = \omega \sqrt{A^2 - x^2}$ — (7)

i.e. $A^2 = \frac{v_0^2}{\omega^2} + x_0^2$ — (8)

From (7),

$$\frac{dx}{dt} = \omega \sqrt{A^2 - x^2}$$

or $\frac{dx}{\sqrt{A^2 - x^2}} = \omega \cdot dt$ — (9)

or, $\int_{x_0}^x \frac{dx}{\sqrt{A^2 - x^2}} = \omega \cdot dt$

or $\left(\sin^{-1} \frac{x}{A} \right)_{x_0}^x = \omega t.$

or $\frac{\sin^{-1} x}{A} - \frac{\sin^{-1} x_0}{A} = \omega t$

or, $\frac{\sin^{-1} x}{A} - \frac{\sin^{-1} x_0}{A} = \omega t$

or $\frac{\sin^{-1} x}{A} - \delta = \omega t$

or $\frac{\sin^{-1} x}{A} = \omega t + \delta$

or $\frac{x}{A} = \sin(\omega t + \delta)$

$\therefore x = A \sin(\omega t + \delta)$

Energy Conservation in Free Oscillation

$$E = ?$$

$$P.E = \frac{1}{2} kx^2$$

$$\therefore P.E = \frac{1}{2} m\omega^2 A^2 \sin^2(\omega t + \delta) \quad \text{--- (1)}$$

$$K.E = \frac{1}{2} mv^2$$

$$\therefore K.E = \frac{1}{2} m A^2 \omega^2 \cos^2(\omega t + \delta) \quad \text{--- (2)}$$

Now,

$$\begin{aligned} \text{Total Energy} &= P.E + K.E \\ \text{or, } E &= \frac{1}{2} m\omega^2 A^2 [\sin^2(\omega t + \delta) + \cos^2(\omega t + \delta)] \end{aligned}$$

$$\therefore E = \frac{1}{2} m\omega^2 A^2$$

#

2) Damped Oscillation.

$$F_{\text{resulting force}} = F_{\text{restoring}} + F_{\text{frictional force}} + F_{\text{external}} \quad \text{--- (1)}$$

$$F_{\text{rest}} = -kx \quad \text{--- (2)}$$

$$F_{\text{frictional}} = -bv \quad \text{--- (3)}$$

$$F_{\text{ext}} = F_D \sin(\omega''t + \delta) \quad \text{--- (4) (driving force)}$$

$$F = -kx - bv + F_D \sin(\omega''t + \delta)$$

$$\text{or, } ma = -kx - bv + F_D \sin(\omega''t + \delta)$$

$$\text{or, } m \cdot \frac{d^2x}{dt^2} = \dots$$

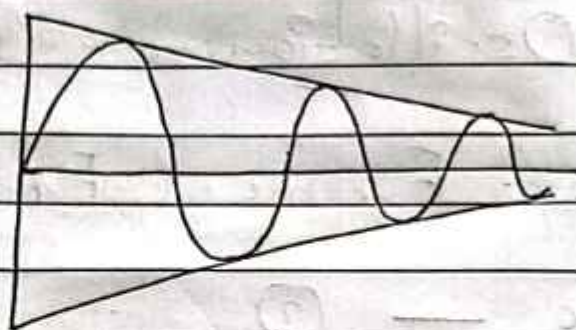
$$F = -kx - bv$$

$$\text{or } ma = -kx - b \cdot \frac{dx}{dt}$$

$$\text{or } m \cdot \frac{d^2x}{dt^2} = -kx - b \cdot \frac{dx}{dt}$$

$$\text{or } \frac{d^2x}{dt^2} + \frac{k}{m}x + \frac{b}{m} \cdot \frac{dx}{dt} = 0$$

$$\text{or } \frac{d^2x}{dt^2} + 2\left(\frac{b}{2m}\right) \frac{dx}{dt} + \frac{k}{m}x = 0 \quad \text{--- (5)}$$



$$x(t) = A \cdot e^{-bt/2m} \cdot \sin(\omega t + \delta) \quad \text{--- (5)}$$

H/W

$$\begin{cases} \frac{dx}{dt} = ? \\ \frac{d^2x}{dt^2} = ? \end{cases}$$

$$\omega' = \sqrt{\omega^2 - \left(\frac{b}{2m}\right)^2}$$

Damped frequency.

$$b = 0$$

$$\boxed{\omega' = \omega}$$

$$\omega > \frac{b}{2m}$$

$$\omega < \frac{b}{2m}$$

Forced Oscillation

$$F = F_{\text{rest}} + F_{\text{fric}} + F_{\text{external (periodic)}}$$

$$F_{\text{rest}} = -kx$$

$$F_{\text{fric}} = -bv$$

$$F_{\text{ext}} = F_D \sin(\omega''t + \delta)$$

$$F = -kx - bv + F_D \sin(\omega''t + \delta)$$

$$m a = -kx - bv + F_D \sin(\omega''t + \delta)$$

$$m \cdot \frac{d^2x}{dt^2} + b \cdot \frac{dx}{dt} + kx = F_D (\sin(\omega''t + \delta))$$

$$m \frac{d^2 x}{dt^2} + \frac{b}{m} \frac{dx}{dt} + \frac{k}{m} x = F_D (\sin(\omega'' t + \delta)) \quad \text{--- (6)}$$

Sol

$$x(t) = A \sin(\omega'' t + \delta) \quad \text{--- (7)}$$

Find

$$\left\{ \begin{array}{l} \frac{dx}{dt} = ? \\ \frac{d^2 x}{dt^2} = ? \end{array} \right.$$

$$A = \pm \frac{F_D}{\sqrt{m^2(\omega''^2 - \omega^2)^2 + b^2 \omega''^2}} \quad \text{--- (7)}$$

$$\omega'' = \omega \Rightarrow$$

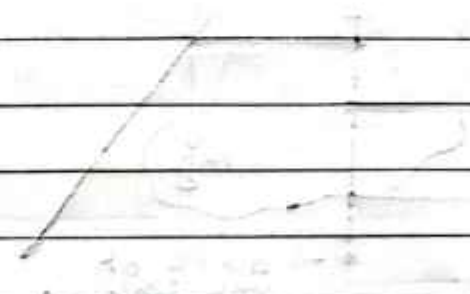
$$A = \pm \frac{F_D}{b \cdot \omega''}$$

$b \rightarrow$ Very Small

$A \rightarrow$ High $\rightarrow$ Resonance

$\downarrow$
Sharpness of resonance.

Q) Find the energy dissipation in damped oscillation?



Energy dissipation = work done by the damping force

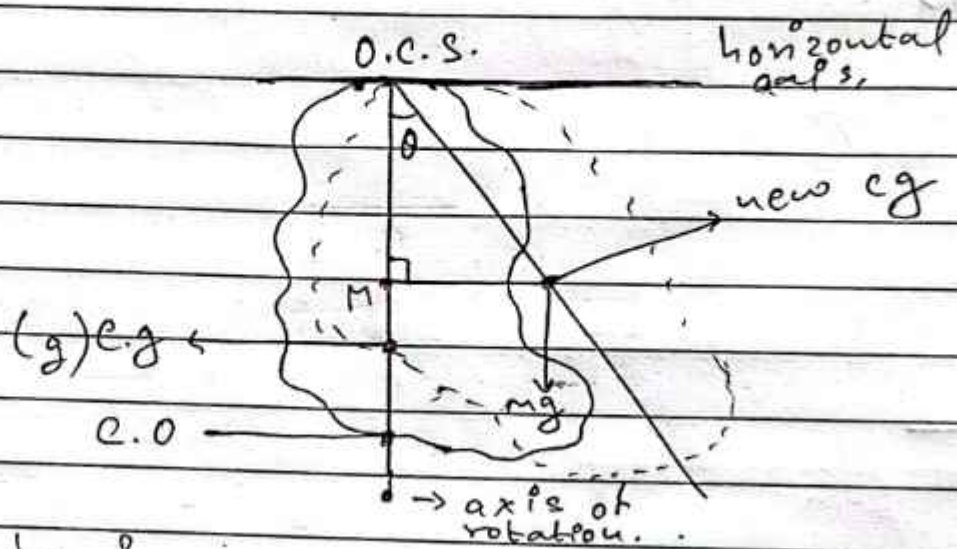
$$dW = F_{\text{damping}} \cdot dx$$

$$F_{\text{damping}} = -r \cdot v$$

$$dW = -r \cdot v \cdot dx$$

$$W = \int_{x_1}^{x_2} -r \cdot v \cdot dx$$

4 Compound Pendulum



$$m g' = ?$$

09 = 2

$$O_1 = 2$$

$$\sin \theta = \frac{g' m}{o g'}$$

$$\therefore mg' = l \sin \theta$$

Restoring torque (τ) = - force $\times$ perpendicular dist. from axis of rotation.

3. $\tau = 3g$

$$\tau = -mg \sin \theta$$

$$\rightarrow I\alpha = -mg\ell\theta$$

$$\Rightarrow \alpha = -\frac{mgl}{I} \theta \quad \text{--- (1)}$$

$\alpha \propto \theta$

$$a \propto x$$

$$\frac{d^2\theta}{dt^2} = -\frac{mgdl}{I}\theta \quad \text{--- (2)}$$

Comparing with eqⁿ,

$$\omega = \sqrt{\frac{mgdl}{I}}$$

$$\frac{d^2\theta}{dt^2} = -\omega^2\theta \quad \text{--- (3)}$$

$$\omega = 2\pi f = \sqrt{\frac{mgdl}{I}}$$

$$T = 2\pi \sqrt{\frac{I}{mgdl}} \quad \text{--- (4)}$$

From (2) & (3),

$$\omega^2 = \frac{mgdl}{I}$$

$$I = I_0 + dl^2M$$

$$I = Mk^2 + Ml^2 \quad \text{--- (5)}$$

In. S. P

$$T = 2\pi \sqrt{\frac{I}{g}}$$

From (4),

$$T = 2\pi \sqrt{\frac{Mk^2 + Ml^2}{Mgdl}}$$

$$\therefore T = 2\pi \sqrt{\frac{k^2 + l^2}{gl}}$$

$$\therefore T = 2\pi \sqrt{\frac{k^2}{g} + l} \quad \text{--- (6)}$$

##

Maximum and Minimum time period of Compound Pendulum

$$T = 2\pi \sqrt{\frac{k^2 + d}{d}} \quad \text{--- (1)}$$

$$d \rightarrow 0$$

$$T \rightarrow \infty$$

$$[T_{\max} = \infty]$$

For minimum,

Squaring eqⁿ (1),

$$T^2 = 4\pi^2 \left[\frac{k^2 + d}{d} \right]$$

Now,

$$\frac{dT^2}{dd} = 4\pi^2 \left[\frac{-k^2 + 1}{d^2} \right]$$

$$\Rightarrow 2T \cdot \frac{dT}{dd} = \frac{4\pi^2}{g} \left[\frac{-k^2 + 1}{d^2} \right]$$

Now,

$$\frac{dT}{dd} = 0$$

$$\therefore \frac{-k^2 + 1}{d^2} = 0$$

$$\Rightarrow k^2 = d^2$$

$$[\therefore k = d]$$

$$\therefore T_{\min} = 2\pi \sqrt{\frac{2l}{g}}$$

Interchangeability of C.S and C.O.

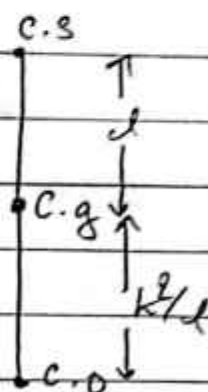
(Centre of Suspension) (Centre of Oscillation).

We have,

$$T = 2\pi \sqrt{\frac{\frac{k^2}{u} + u}{g}} \quad \text{--- (1)}$$

$$\text{Let, } \frac{k^2}{u} = u' \quad \text{--- (2)}$$

$$k^2 = uu' \quad \text{--- (3)}$$



From (1) & (3),

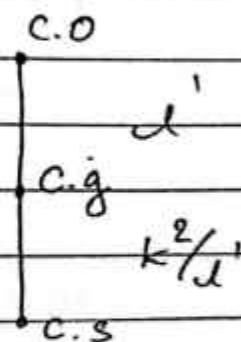
$$T = 2\pi \sqrt{\frac{uu' + u}{g}}$$

$$\text{or, } T = 2\pi \sqrt{\frac{u' + u}{g}} \quad \text{--- (4)}$$

After inversion,

$$T' = 2\pi \sqrt{\frac{k^2 + u'}{u'}}$$

$$\text{or, } T' = 2\pi \sqrt{\frac{uu' + u'}{u'}}$$



$$\therefore T' = 2\pi \sqrt{\frac{u + u'}{g}} \quad \text{--- (5)}$$

classmate
Date _____
Page _____

Determination of acceleration due to gravity 'g'.

We have,

$$T = 2\pi \sqrt{\frac{k^2 + l}{g}} \quad \text{--- (1)}$$

$$\Rightarrow T^2 = \frac{4\pi^2}{g} \left[\frac{k^2 + l}{1} \right]$$

$$\Rightarrow T^2 = \frac{4\pi^2 l}{g} + \frac{4\pi^2 k^2}{g} \quad \text{--- (2)}$$

Multiplying by l ,

$$T^2 l = \frac{4\pi^2 l^2}{g} + \frac{4\pi^2 k^2 l}{g} \quad \text{--- (2)}$$

Comparing with, $y = mx + c$.

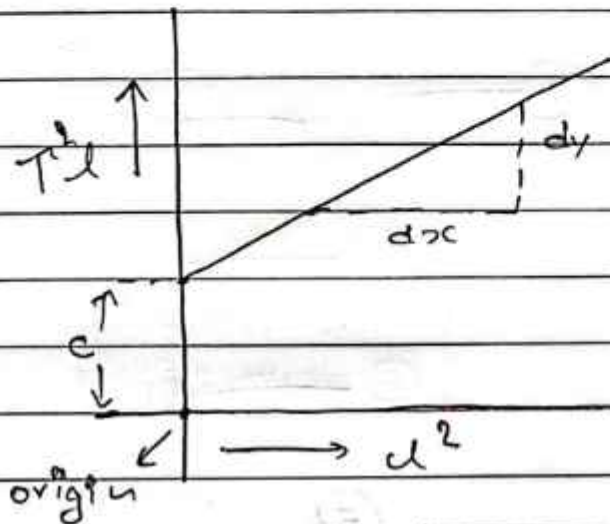
$$y = T^2 l, \quad m = \frac{4\pi^2}{g}, \quad x = l^2, \quad c = \frac{4\pi^2 k^2}{g}$$

$m = \text{slope}$

$$\therefore \text{Slope} = \frac{4\pi^2}{g} \quad \text{--- (3)}$$

$$\text{i.e. } \frac{dy}{dx} = \frac{4x^2}{g} \quad \text{--- (4)}$$

$$\therefore g = \frac{4\pi^2}{\left(\frac{dy}{dx}\right)} \quad \text{--- (5)}$$



* Torsional Pendulum

θ = twisting angle

τ = twisting force

$$\tau \propto \theta$$

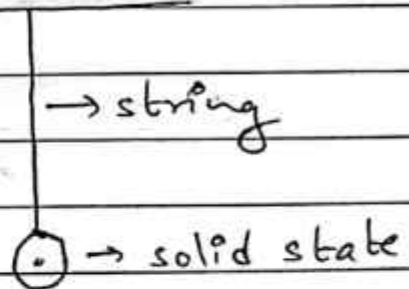
$$\text{or } \tau = -c\theta \quad \text{--- (1)}$$

$$c = -\frac{\tau}{\theta} \rightarrow \text{couple per unit twist}$$

$$\Rightarrow \tau = I\alpha \quad \text{--- (2)}$$

$$\therefore I\alpha = -c\theta$$

$$\Rightarrow \alpha = -\frac{c}{I}\theta \quad \text{--- (3)}$$



Comparing with,

$$\alpha = -\omega^2 \theta \quad \text{--- (4)}$$

$$\therefore \omega^2 = \frac{c}{I}$$

$$\Rightarrow (2\pi f)^2 = \frac{c}{I}$$

$$\Rightarrow 2\pi f = \sqrt{\frac{c}{I}}$$

$$\Rightarrow \frac{2\pi}{T} = \sqrt{\frac{c}{I}}$$

$$\Rightarrow T = 2\pi \sqrt{\frac{I}{c}} \quad \text{--- (5)}$$

$I = MI$ of disc (Moment of Inertia)

$$\therefore I = \frac{1}{2} MR^2 \quad \text{--- (6)}$$

Also,

$M = \text{mass of disc}$

$R = \text{radius of disc}$

$$c = \frac{\pi \eta r^4}{2l} \quad \text{--- (7)}$$

$\eta = \text{modulus of rigidity of string}$

In eqⁿ (5),

$$T = 2\pi \sqrt{\frac{\frac{1}{2} MR^2 \cdot 2l}{\pi \eta r^4}}$$

$$\text{or, } T^2 = \frac{4\pi^2 MR^2 l}{\pi \eta r^4}$$

$$\therefore \eta = \frac{4\pi MR^2 l}{T^2 r^4}$$

Numerical

- Q) A particle executing Simple Harmonic Motion of amplitude 'A' along the x-axis. At $t=0$, the position of the particle is $x = A/2$. Find the phase constant if,
 $x = A \sin(\omega t + \phi)$.

Soln

$$\phi = ?$$

$$x = A \sin(\omega t + \phi)$$

$$A = A \sin(\omega \times 0 + \phi)$$

$$\frac{1}{2} = \sin(0 + \phi)$$

$$\frac{1}{2} = \sin 0 + \sin \phi$$

$$\frac{1}{2} = \sin \phi$$

$$\therefore \phi = 30^\circ$$

Q) At a time when the displacement is half of the amplitude, what fraction of total energy is kinetic and what fraction is potential in SHM? At what displacement the potential and kinetic energy are halved.

$$TE = \frac{1}{2} M \omega^2 A^2, \quad PE = \frac{1}{2} M \omega^2 x^2, \quad KE = \frac{1}{2} M \omega^2 (A^2 - x^2).$$

Soln

When, $x = A/2$

Then

$$KE = \frac{1}{2} M \omega^2 \left(A^2 - \frac{A^2}{4} \right)$$

$$= \frac{1}{2} M \omega^2 \frac{3A^2}{4}$$

$$= \frac{3}{4} \times \frac{1}{2} M \omega^2 A^2$$

$$\therefore KE = \frac{3}{4} TE$$

Again,

$$PE = \frac{1}{2} M \omega^2 x^2$$

$$= \frac{1}{2} M \omega^2 \frac{A^2}{4}$$

$$= \frac{1}{4} \times \frac{1}{2} M \omega^2 A^2$$

$$\therefore PE = \frac{1}{4} TE$$

Now,

$$KE = PE$$

$$\frac{1}{2} M \omega^2 A^2 - \frac{1}{2} M \omega^2 x^2 = \frac{1}{2} M \omega^2 x^2$$

$$\therefore \frac{1}{2} M \omega^2 A^2 = M \omega^2 x^2$$

$$\therefore \frac{A^2}{2} = x^2$$

$$\therefore x = \frac{A}{\sqrt{2}}$$

✓

- 1) An oscillator of mass 10 kg has a velocity of 5 m/s after 1 sec of it crossing the mean position. If the time period is 5 sec, find the kinetic and potential energy at that instant.

Soln

$$m = 10 \text{ kg}$$

$$PE = ?$$

$$v = 5 \text{ m/s}$$

$$KE = ?$$

$$t = 1 \text{ sec}$$

$$T = 5 \text{ sec}$$

Then,

$$KE = \frac{1}{2} mv^2 = \frac{1}{2} \times 10 \times 5^2 = 125$$

Now,

$$v = A\omega \cos \omega t$$

$$\therefore 5 = A \times 1.25 \cos(1.25 \times 1)$$

$$\therefore 4 = A \cdot 0.99$$

$$\therefore A = 4$$

$$\begin{aligned} \omega &= 2\pi f \\ &= 2\pi/T \\ &= 2\pi/5 = 1.25 \end{aligned}$$

Here,

$$\begin{aligned} x &= A \sin \omega t \\ &= 4 \sin(1.25 \times 1) \\ &= 0.087 \end{aligned}$$

$$PE = \frac{1}{2} m \omega^2 x^2$$

$$= \frac{1}{2} \times 10 \times 1.25^2 \times 0.087^2 = 0.05$$

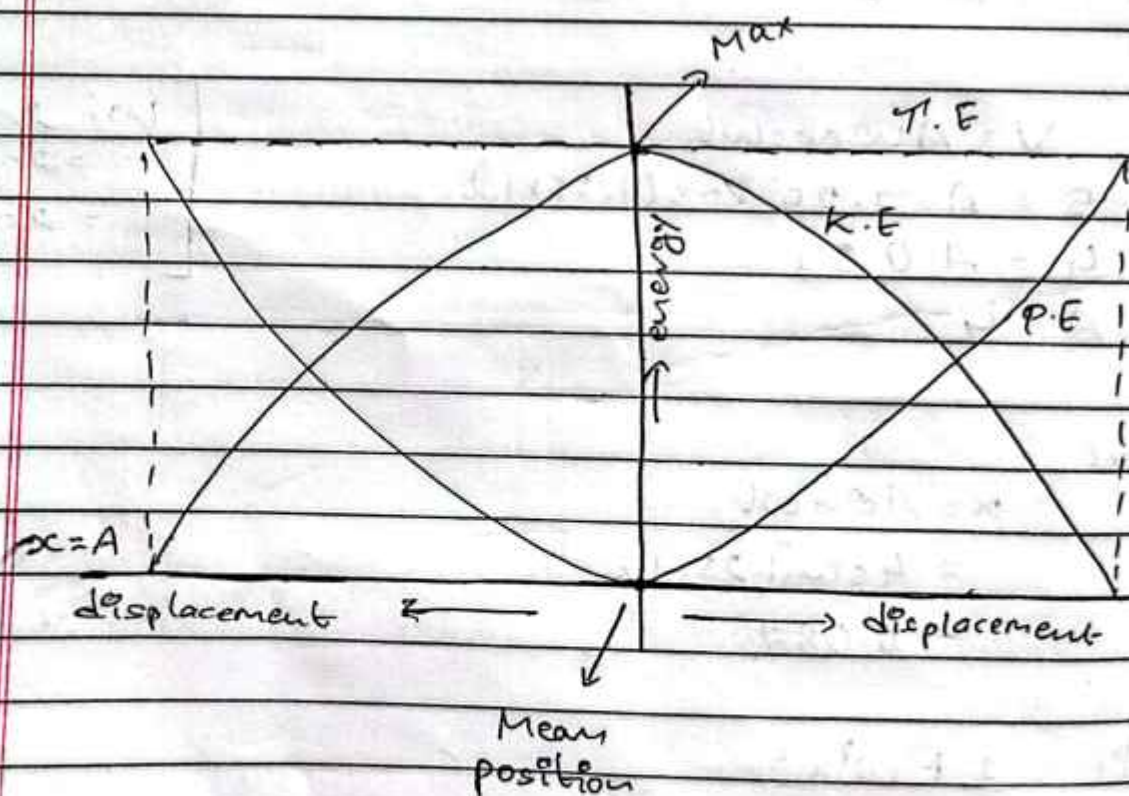
2) Plot KE, PE and total energy graphically.

$$PE = \frac{1}{2} kx^2$$

$$\therefore PE = \frac{1}{2} M\omega^2 x^2$$

$$\therefore KE = \frac{1}{2} M\omega^2 (A^2 - x^2)$$

$$\therefore TE = \frac{1}{2} M\omega^2 A^2$$



displacement of a particle
executing harmonic motion.



3) If $x = A \sin(\omega t + \phi)$ find the expression for,

i) Max displacement = ?

ii) Max velocity = ?

iii) $a_{\text{cl}}^{\text{m}} = ?$

iv) Max $a_{\text{cl}}^{\text{m}} = ?$

v) Max force = ?

Soln

i) $\sin(\omega t + \phi) = \pm 1$

$$\therefore x = A(\pm 1)$$

$$\boxed{\therefore x = \pm A}$$

ii) $\frac{dx}{dt} = A \cos(\omega t + \phi) \cdot \omega$

$$v = A\omega \cos(\omega t + \phi)$$

v will be max^{m} when, $\cos(\omega t + \phi) = \pm 1$.

$$v = A\omega(\pm 1)$$

$$\boxed{\therefore v = \pm A\omega}$$

iii) $\frac{d^2x}{dt^2} = -A\omega^2 \sin(\omega t + \phi)$

$$a = -A\omega^2 \sin(\omega t + \phi)$$

$$\sin(\omega t + \phi) = \pm 1$$

iv) $\text{max} \quad \boxed{\therefore a = \pm A\omega^2}$

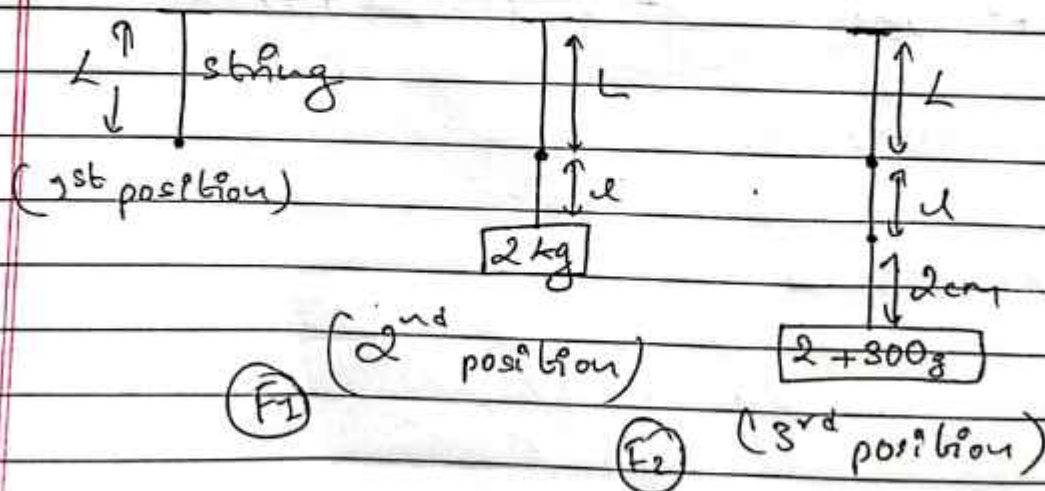
v) Force, $F = ma$
 $= m(-A\omega^2 \sin(\omega t + \delta))$

Max^m, $\sin(\omega t + \delta) = \pm 1$

$F = M(+A\omega^2)$

$\therefore F_{\text{max}} = \pm A\omega^2 M$

- Q) A 2 kg mass hang from a spring. A 200 g body hang below the mass stretches the string 2 cm further. If the mass ~~300g~~ 300g is removed and the remaining mass set into oscillation, find the period of the oscillation?



In 2nd position,

$$F_2 = -kx \quad \text{--- (1)}$$

$$F_2 = -k(l+2) \quad \text{--- (2)}$$

Resultant force,

$$F = F_2 - F_1$$

$$\text{or, } F = -k(l+2) + kl$$

$$\text{or, } F = -k \times 2 \quad \text{--- (3)}$$

$$\text{or } mg = -kx$$

$$\text{or } 300 \times 10^{-3} \times 10 = -k \times 2 \times 10^{-2}$$

$$\text{or, } k = -150$$

$$\therefore k = 150 \quad \text{--- (4)}$$

Now,

$$T = \frac{2\pi}{\omega} \quad \text{--- (5)}$$

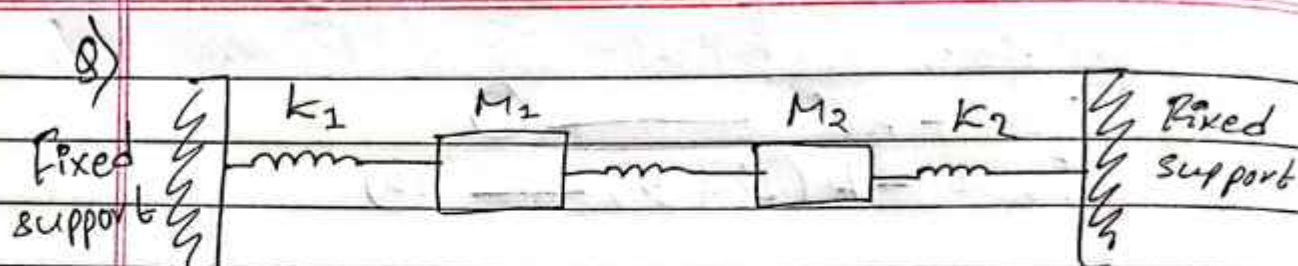
Also,

$$\omega = \sqrt{\frac{k}{M}}$$

$$\therefore T = 2\pi \sqrt{\frac{M}{k}}$$

$$= 2\pi \sqrt{\frac{2}{150}}$$

$$= 0.72$$



Find the frequency of oscillation?

$$\left\{ \begin{array}{l} F_1 = ? \\ F_2 = ? \end{array} \right. , \left\{ \begin{array}{l} -F_1 = -k_1 x \\ F_2 = -k_2 x \end{array} \right.$$

$$\therefore F = F_1 + F_2$$

$$\therefore F = -(k_1 + k_2)x \quad \text{--- (1)}$$

$$F = -\omega^2 x \quad \text{--- (2)}$$

From (1) & (2),

$$-\omega^2 x = -(k_1 + k_2)x$$

$$[F = -kx] \quad \text{--- (3)}$$

$$K = k_1 + k_2$$

$$M\omega^2 = k_1 + k_2$$

(M = combined mass)

$$\text{or } \omega^2 = \frac{k_1 + k_2}{M}$$

$$\text{or } \omega = \sqrt{\frac{k_1 + k_2}{M}}$$

$$\therefore f = \frac{1}{2\pi} \sqrt{\frac{k_1 + k_2}{M}}$$

- Q) A disc 1m in diameter is cut from a metal sheet. The disc is made to swing as a pendulum by drilling a small hole in it and mounting it on a nail driven into a wall. If 'x' be the distance from the nail to the centre of plate.
- for what values of x, the period becomes 1.7 sec.
 - for what value of x, the time period is min?

Soln

$$T = 2\pi \sqrt{\frac{k^2/x + x}{g}} \quad \text{--- (1)}$$

$$\text{Moment of Inertia (MI) of disc} = \frac{1}{2} MR^2 \quad \text{--- (2)}$$

$$\text{MI of disc in terms of } k = Mk^2 \quad \text{--- (3)}$$

$$\frac{1}{2} MR^2 = Mk^2$$

$$\therefore R^2 = 2k^2$$

$$k = \frac{R}{\sqrt{2}} \quad \text{--- (4)}$$

From (1) & (4), we get

$$T_{\min} = 2\pi \sqrt{\frac{2k}{g}}$$

$$\therefore T_{\min} = 2\pi \sqrt{\frac{2 \times R/\sqrt{2}}{g}}$$

$$\therefore T_{\min} = 1.6k$$

$$k = 2R$$

Now,

a) From (1), $T = 2.7 \text{ s.}$

$$T = 2\pi \sqrt{\frac{k^2 + x}{g}}$$

$$\text{or } 2.7 = 2\pi \sqrt{\frac{R^2 + x^2}{2gx}}$$

$$\text{or } 2.89 = 4\pi^2 \left(\frac{R^2 + x^2}{2gx} \right)$$

$$\text{or } 0.07 \times 2gx = R^2 + x^2$$

$$\text{or } 1.4x = R^2 + x^2 \quad (R = \text{radius})$$

$$\text{or } 1.4x - x^2 = \frac{1}{4} \quad (\text{Diam} = 1)$$

$$(R = \frac{1}{2})$$

$$\text{or } 1.4 - x^2 - \frac{1}{4} = 0$$

$$\therefore x = 1.18 \quad \text{or} \quad x = 0.21$$

So, x can be 1.18 or 0.21

Now,

b) For Min^m Period,

$$T_{\min} = 2\pi \sqrt{\frac{2k}{g}}$$

$$= 2\pi \sqrt{\frac{2 \times R/\sqrt{2}}{g}}$$

$$= 2\pi \sqrt{\frac{R\sqrt{2}}{g}}$$

$$= 2\pi \sqrt{\frac{1 \cdot \sqrt{2}}{2g}}$$

$$(g = 9.8)$$

$$= 1.67 \text{ sec}$$

Unit - 2

02/16

Wave Motion.

Mechanical
Wave
(Sound wave)

Electromagnetic
Wave
(Rays)

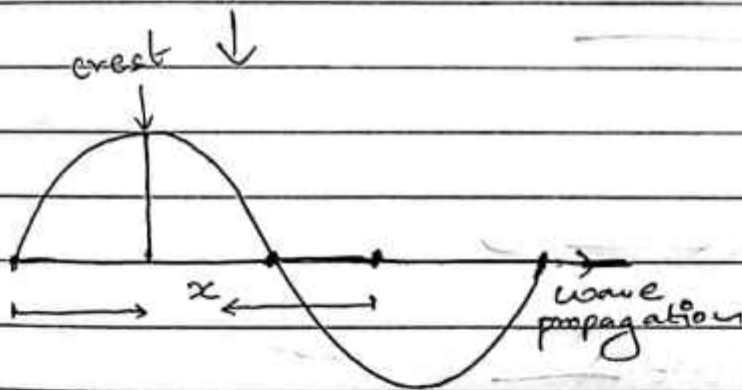
Matter
Wave
(e^- , p^+)

Gravitational
Wave.

Mechanical Wave

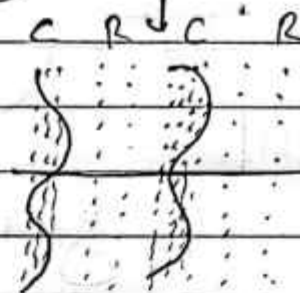
(on the basis of vibration of particles)

Transverse



λ (wavelength)
 2π (phase)

Longitudinal



When path diff (Δ) $\rightarrow$ then phase diff (2π)

" 1 $\rightarrow$ " 2π

" x $\rightarrow$ " $\frac{2\pi x}{\lambda}$

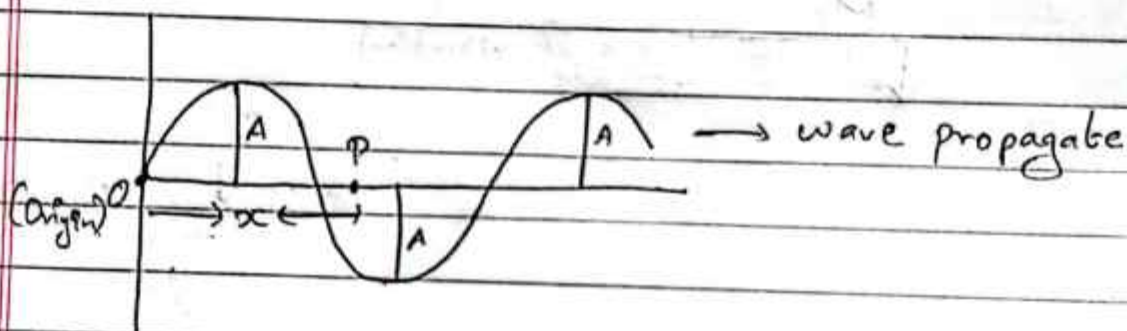
$$\text{phase diff } (\phi) = \frac{2\pi}{\lambda} \times \text{path diff } (x)$$

* On the basis of transportation of energy

Travelling Wave
(Progressive)

Stationary Wave
(Standing)

* Travelling Wave



$$y = f(x, t) \text{ — (1)}$$

$$\Rightarrow f(x, t) = A \sin \omega t \text{ — (2)}$$

$$\therefore y = A \sin \omega t \text{ — (3)}$$

If 'P' be the point at a distance 'x' from 'O', then displacement of particle at point P. i.e.

$$y = A \sin(\omega t - \phi) \text{ — (4)}$$

↓
phase lagging.

When, path diff $\lambda \rightarrow$ phase diff 2π

$$1 \rightarrow \text{"} \text{"} \frac{2\pi}{\lambda}$$

$$x \rightarrow \text{"} \left(\frac{2\pi}{\lambda} \cdot x \right) \Rightarrow \phi$$

$$\therefore \phi = \frac{2\pi}{\lambda} x \rightarrow (5)$$

$\therefore$ From (4) & (5),

$$y = A \sin(\omega t - \frac{2\pi}{\lambda} x)$$

$$\therefore y = A \sin(\omega t - kx)$$

$$[\because 2\pi/\lambda = k]$$

$\downarrow$
wave number
i.e. constant.

$$\therefore y = A \sin(\omega t + kx)$$

$\hookrightarrow$ for wave travelling along negative x-direction

$\therefore$ In general,

$$\therefore y = A \sin(\omega t \pm kx)$$

* phase (Wave) velocity and particle ~~displ~~ velocity.

$v \rightarrow$ particle velocity

We have,

$$y = A \sin(\omega t - kx) \quad \text{--- (1)}$$

$$\text{or, } \frac{dy}{dt} = A\omega \cos(\omega t - kx) \cdot$$

$$\text{or, } v = A\omega \cos(\omega t - kx) \quad \text{--- (2)}$$

Also,

$$\frac{dy}{dx} = -kA \cos(\omega t - kx) \quad \text{--- (3)}$$

Also, We have,

$\omega t - kx = \text{phase} = \text{constant}$
Diff. w.r.t. t ,

$$\omega - k \cdot \frac{dx}{dt} = 0$$

$$\Rightarrow \frac{dx}{dt} = \frac{\omega}{k}$$

$$\Rightarrow u = \frac{\omega}{k} \quad \text{--- (4)}$$

$$\Rightarrow u = \frac{2\pi f}{2\pi/\lambda}$$

$$\Rightarrow u = f\lambda$$

($u \rightarrow$ phase (Wave) velocity).

From eqⁿ ② & ③

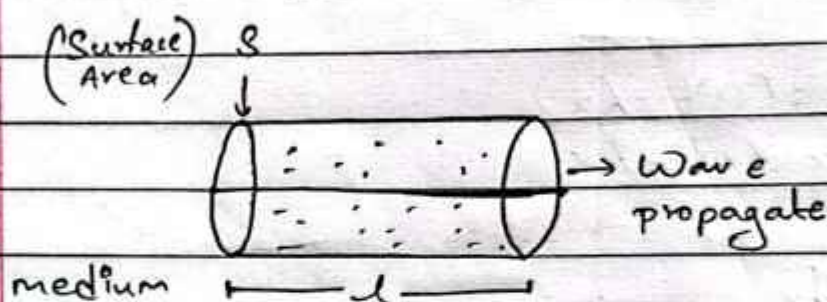
$$\frac{v}{dy/dx} = \frac{A\omega \cos(\omega t - kx)}{-Ak \cos(\omega t - kx)}$$

$$\Rightarrow v = - \left(\frac{\omega}{k} \right) \frac{dy}{dx}$$

$$\therefore v = u \left(- \frac{dy}{dx} \right)$$

{ Particle velocity = Wave velocity $\times$ -ve slope }

Intensity of Wave



$$\text{Volume of medium} = Sl \quad \text{--- (1)}$$

$$\text{let, } n = \frac{\text{no. of particles}}{\text{volume}}$$

$$\therefore \text{no. of particles} = n \times \text{volume}$$

$$\therefore \text{Total no. of particles} = nSl \quad \text{--- (2)}$$

We have,

$$\text{Energy of a particle} = \frac{1}{2} Mw^2 A^2 \quad \text{--- (3)}$$

Total energy of particles,

$$E = \frac{1}{2} Mw^2 A^2 \times nSl \quad \text{--- (4)}$$

If 'I' be the intensity,

$$I = \frac{\text{Energy}}{\text{Area} \times \text{Sec}}$$

$$\Rightarrow I = \frac{\frac{1}{2} Mw^2 A^2 nSl}{St}$$

$$\Rightarrow I = \frac{1}{2} (\mu) \omega^2 A^2 \left(\frac{d}{t} \right)$$

$$\Rightarrow I = \frac{1}{2} \rho \omega^2 A^2$$

$$\Rightarrow I = \frac{1}{2} 4 \pi^2 f^2 A^2 \mu$$

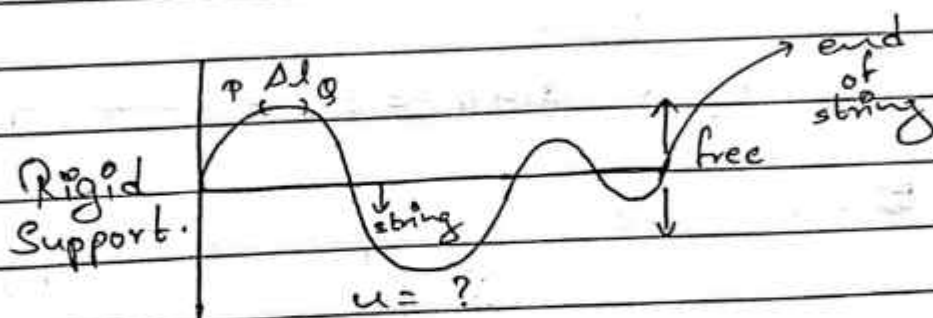
$$\Rightarrow \boxed{I = 2 \pi^2 f^2 A^2 \mu}$$

$$I \propto A^2$$

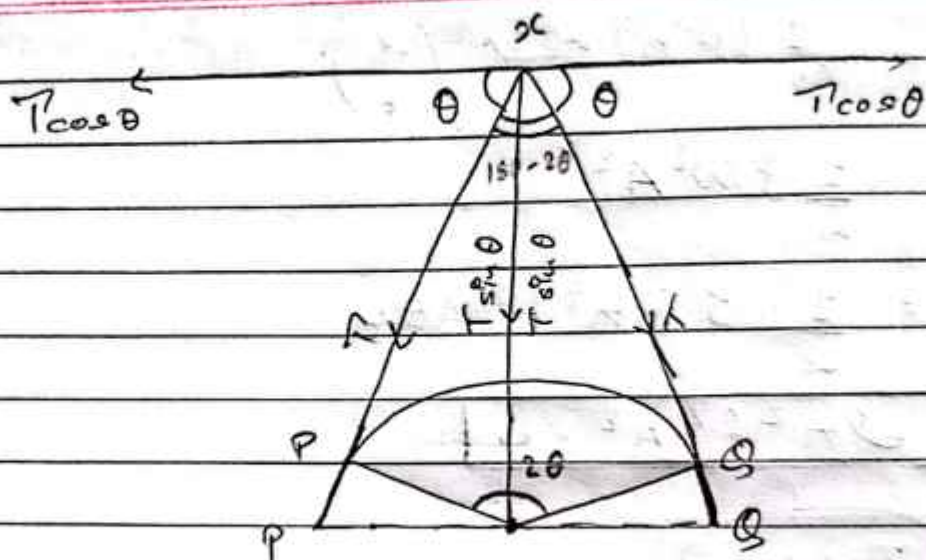
$$I = \frac{\text{energy}}{\text{Area} \times \text{time}}$$

$$P = I \times \text{Area.}$$

Velocity of transverse wave along a Stretched String :-



$$PQ = \Delta l = \text{element length.}$$



$$\text{Resultant tension} = 2T \sin \theta \quad \text{--- (1)}$$

↓
Centripetal force

$$\text{Also, centripetal force} = \frac{mu^2}{R} \quad \text{--- (2)}$$

From (1) & (2),

$$\frac{mu^2}{R} = 2T \sin \theta \quad \text{--- (3)}$$

$$\text{for small angle, } \sin \theta \approx \theta. \quad \text{--- (4)}$$

If μ be the mass per unit length, then,

$$\mu = \frac{\text{mass } (m)}{\text{length } (\Delta l)}$$

$$\text{or } m = \mu \cdot \Delta l \quad \text{--- (5)}$$

Now,

$$\frac{m \Delta l \cdot u^2}{R} = 2\theta T \quad \text{--- (6)}$$

ΔPOQ ,

Central angle = $\frac{\text{Arc}}{\text{radius}}$

$$\text{or, } 2\theta = \frac{\Delta l}{R} \quad \text{--- (7)}$$

From (6) & (7),

$$\frac{m \cdot \Delta l \cdot u^2}{R} = \frac{\Delta l}{R} \cdot T$$

$$\text{or } m \cdot u^2 = T$$

$$\text{or } u^2 = \frac{T}{m}$$

$$\therefore u = \sqrt{\frac{T}{m}}$$

* Energy transmission along a stretched string

$$\frac{dE}{dt} = ?$$

$$\text{power} = ?$$

$$E = KE + P.E \quad \text{--- (1)}$$

$$KE = \frac{1}{2} mv^2 \quad \text{--- (2)}$$

$$PE = \frac{1}{2} m\omega^2 y^2 \quad \text{--- (3)}$$

$$m = \mu \Delta x \quad \text{--- (4)}$$

$$y = A \sin(\omega t - kx) \quad \text{--- (5)}$$

$$v = A\omega \cos(\omega t - kx) \quad \text{--- (6)}$$

Now,

$$KE = \frac{1}{2} \mu \cdot \Delta x \cdot A^2 \omega^2 \cos^2(\omega t - kx)$$

$$\frac{KE}{t} = \frac{1}{2} \mu \cdot u \cdot A^2 \omega^2 \cos^2(\omega t - kx) \quad \text{--- (7)}$$

Again,

$$PE = \frac{1}{2} \mu \cdot \Delta x \cdot \omega^2 \cdot A^2 \sin^2(\omega t - kx)$$

$$\frac{PE}{t} = \frac{1}{2} \mu \cdot u \cdot \omega^2 A^2 \sin^2(\omega t - kx) \quad \text{--- (8)}$$

Here,

From (7) & (8),

Total rate of change of energy,

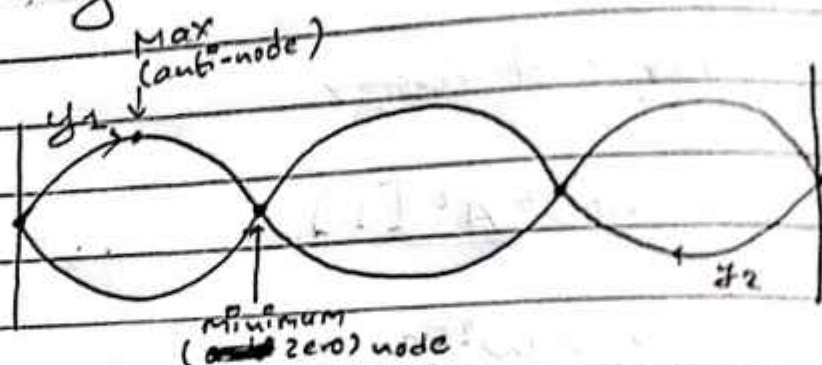
$$\frac{E}{t} = \frac{1}{2} \mu \cdot u \cdot \omega^2 \cdot A^2 \quad [1]$$

$$\Rightarrow \text{Power} = \frac{1}{2} \mu \cdot u \omega^2 A^2$$

$$\Rightarrow P = \frac{1}{2} \mu \cdot u \cdot 4\pi^2 f^2 A^2$$

$$\Rightarrow \underline{P = 2\pi^2 f^2 A^2 \mu \cdot u}$$

Standing Wave (Stationary Wave)



$$y_1 = A \sin(\omega t - kx) \text{ ————— (1)}$$

$$y_2 = A \sin(\omega t + kx) \text{ ————— (2)}$$

$$y = y_1 + y_2 \text{ ————— (3)}$$

$$y = A [\sin(\omega t - kx) + \sin(\omega t + kx)]$$

$$\Rightarrow y = A \cdot \frac{2 \sin(\omega t - kx + \omega t + kx)}{2} \cdot \frac{\cos(\omega t - kx - \omega t - kx)}{2}$$

$$\Rightarrow y = 2A \sin \omega t \cdot \cos kx$$

$$\Rightarrow y = (2A \cos kx) \sin \omega t \text{ ————— (4)}$$

Generally,

$$y = A \sin \omega t \text{ ————— (5)}$$

$$2A \cos kx = B \text{ ————— (6)}$$

↓
amplitude of
stationary wave.

From (4) & (6),

$$y = B \sin \omega t \quad \text{--- (7)}$$

Condition for node and anti-node,

I) For Node :

$$B = 0$$

$$\cos kx = 0$$

$$\cos kx = \cos \frac{n\pi}{2} \quad [n = 1, 3, 5, \dots]$$

$$\Rightarrow kx = \frac{n\pi}{2}$$

$$\Rightarrow x = \frac{n\pi}{2k}$$

$$\Rightarrow x = \frac{n\pi}{2 \cdot 2\pi/\lambda}$$

$$\Rightarrow x = \frac{n\lambda}{4}$$

$$\Rightarrow x = \frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}, \dots$$

These are the positions of node.

→

II) For Anti-node

$$B = \pm 1$$

$$\Rightarrow \cos kx = \pm 1$$

$$\Rightarrow \cos kx = \cos n\pi, [n = 0, 1, 2, \dots]$$

$$\Rightarrow kx = n\pi$$

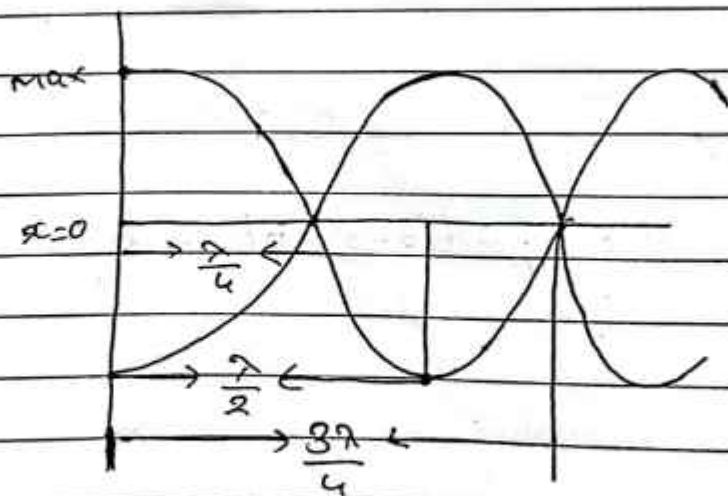
$$\Rightarrow x = \frac{n\pi}{k}$$

$$\Rightarrow x = \frac{n\pi \cdot \lambda}{2\pi}$$

$$\Rightarrow x = \frac{n \cdot \lambda}{2}$$

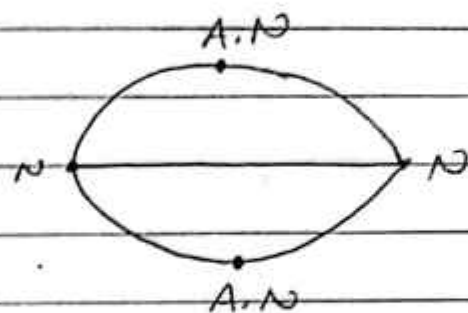
$$\Rightarrow x = 0, \frac{\lambda}{2}, \lambda, \dots$$

$\therefore$ These are the position of anti-node.



nodes & antinodes formed alternatively.

Resonance of Stationary Wave.



Fundamental
Frequency
(f_0)

$$\begin{aligned} \text{string length} &= l \\ \text{wave length} &= \frac{\lambda}{2} \end{aligned}$$

$$\Rightarrow l = \frac{\lambda}{2}, \text{ when only one antinode is formed.}$$

When,

$$\begin{aligned} u &= f_0 \lambda \\ \Rightarrow f_0 &= \frac{u}{\lambda} \end{aligned}$$

$$\Rightarrow f_0 = \frac{u}{2l}$$

↓

frequency required
for 1st harmonic

$$\therefore [f_n = (n+1) f_0]$$

Q) A wave of frequency 500 cycles per sec has wave velocity of 350 m/s. How apart are two points 60° out of phase?

Soln

$$f = 500 \text{ cycles/sec}$$

$$u = 350 \text{ m/s}$$

$$x = ? \text{ (path diff)}$$

$$\text{phase diff } (\phi) = 60^\circ$$

Now,

$$u = f\lambda$$

$$\therefore 350 = 500 \cdot \lambda$$

$$\therefore \frac{350}{500} = \lambda$$

$$\therefore \frac{7}{10} = \lambda$$

$$\therefore \lambda = 0.7$$

Here,

$$\phi = \frac{\dots \times \pi}{180} = \dots \text{ radian}$$

$$\phi = \frac{2\pi \times x}{\lambda}$$

$$\therefore \frac{60 \times \pi}{180} = \frac{2\pi \times x}{0.7}$$

$$\therefore \frac{60 \times \pi \times 0.7}{180 \times 2\pi} = x$$

$$\therefore x = 0.11$$

Q) The equation of transverse wave travelling in a rope is given by $y = -10 \sin[0.01x - 2t]$. Find, amplitude, wave velocity, max. particle velocity, wavelength and frequency.

Solⁿ

$$y = -10 \sin[0.01x - 2t]$$

On comparing to, $y = A \sin(\omega t - kx)$, we get,

$$i) A = -10$$

$$\omega = -2$$

$$k = -0.01$$

Now,

$$k = \frac{2\pi}{\lambda}$$

$$-0.01 = \frac{2\pi}{\lambda}$$

$$\therefore \lambda = -628.31$$

Here,

$$\omega = 2\pi f$$

$$-2 = 2\pi f$$

$$\therefore f = -0.31$$

Also,

$$\cancel{628.31} \quad u = f \lambda$$

$$= -0.31 \times -628.31$$

$$= 194.7761$$

Again,

$$V_{\max} = A \omega$$

$$= -10 \times -2$$

$$= 20$$

Q) A stretched string has a linear mass density of 5.0 g/cm and a tension of 20 N . A wave on this string has an amplitude of 0.12 mm and frequency of 100 Hz is travelling along negative x direction. Write wave eqⁿ with appropriate units.

Solⁿ

$$\mu = 5 \text{ g/cm} = 0.5 \text{ kg/m}$$

$$T = 20 \text{ N}$$

$$A = 0.12 \text{ mm} = 0.012 \times 10^{-3}$$

$$f = 100 \text{ Hz}$$

$$y = A \sin(\omega t + kx)$$

$$u = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{20}{0.5}} = 4.47 \text{ m/s}$$

$$u = f\lambda$$

$$\therefore 4.47 = 100 \times \lambda$$

$$\therefore \lambda = 0.0447$$

Here,

$$k = \frac{2\pi}{\lambda}$$

$$= \frac{2\pi}{0.0447}$$

$$= 140.56$$

$$\begin{aligned}\omega &= 2\pi f \\ &= 2\pi \cdot 100 \\ &= 200\pi \\ &= 628.31\end{aligned}$$

So,

$$y = 0.012 \times 10^{-3} \sin(200\pi t + 140.56 \cdot x)$$

π

- classmate
Date _____
Page _____
- Q) A string of mass $2g$ for unit length carries progressive wave of amplitude $1.5cm$, frequency $60s^{-1}$ and speed $20m/s$. Calculate
- Energy per meter length of the wire?
 - The rate of energy propagation in the wave?

Soln

$$\begin{aligned}\text{Mass (m)} &= 2g = 2 \times 10^{-3} \text{ kg} \\ \text{Amplitude (A)} &= 1.5cm = 0.015m \\ \text{Frequency (f)} &= 60s^{-1} \\ u &= 20m/s.\end{aligned}$$

Now,

$$\begin{aligned}a) \quad E &= \frac{1}{2} m \omega^2 A^2 \\ &= \frac{1}{2} \times (2 \times 10^{-3}) \times (2\pi f)^2 \times (0.015)^2 \\ &\quad \text{length} \\ &= \frac{1}{2} \times 2 \times 10^{-3} \times (2\pi \times 60)^2 \times (0.015)^2 \\ &= 0.031 \text{ J}\end{aligned}$$

$$b) \quad E = \frac{1}{2} \mu \omega^2 A^2 u$$

$$0.031 = \frac{1}{2} \times \mu \cdot (2\pi \times 60)^2 \times (0.015)^2 \times 20$$

$$\therefore \mu = 9.69 \times 10^{-5}$$

Q) Two travelling waves of equal amplitude and frequencies moves in opposite direction along a string. They interfere to produce standing wave having eqⁿ $y = A \cos kx \sin \omega t$ in which $A = 1 \text{ mm}$, $\omega = 78.5 \text{ s}^{-1}$ and $k = 1.57 \text{ cm}^{-1}$

Find:

- Velocity of travelling wave
- Position of node, closest to origin, $x > 0$.
- Position of antinode closest to origin, $x > 0$.
- Amplitude of particle at $x = 2.33 \text{ cm}$.

Solⁿ

a) $u = f \lambda$

$$u = \frac{\omega}{k} = \frac{78.5}{1.57} = 50 \text{ cm}^{-1} \text{ s}^{-1}$$

b) $\cos kx = 0$

$$\cos kx = \cos \frac{n\pi}{2} \quad [n = 1, 3, 5, \dots]$$

$$kx = \frac{\pi}{2} \Rightarrow x = \frac{\pi}{2k}$$

$$= \frac{3.1415}{2 \times 1.57}$$

$$= 1 \text{ cm}$$

c) $\cos kx = +1$

$$\cos kx = \cos n\pi \quad [\because n = 0, 1, 2, \dots]$$

$$\therefore kx = \pi$$

$$\therefore x = \frac{\pi}{k}$$

$$n x = \frac{\pi}{1.57}$$

$$\therefore x = 2$$

(mm $\rightarrow$ cm $\rightarrow$... / 10)

d) $B = ?$

We know,

$$B = A \cos kx$$

$$= \frac{1}{10} \times \cos (1.57 \times 2.33)$$

$$= 0.099 \quad \nearrow$$

Q) Show that the particle acceleration in progressive wave is equal to the product of square of wave velocity and curvature of displacement.

Soln

$$\frac{d^2y}{dt^2} = u^2 \frac{d^2y}{dx^2} \quad \text{--- (1)}$$

Now,

For progressive wave,

$$y = A \sin(\omega t - kx)$$

Diff. w.r. t. t ,

$$\frac{dy}{dt} = A\omega \cos(\omega t - kx)$$

$$\frac{d^2y}{dt^2} = -A\omega^2 \sin(\omega t - kx) \quad \text{--- (2)}$$

Again, with, r. t. x ,

$$\frac{dy}{dx} = A\omega \cos(\omega t - kx) = A(-k) \cos(\omega t - kx)$$

$$\frac{d^2y}{dx^2} = -Ak^2 \sin(\omega t - kx) \quad \text{--- (3)}$$

Equating eqⁿ (2) & (3) in (1), we get,

$$-A\omega^2 \sin(\omega t - kx) = u^2 (-Ak^2 \sin(\omega t - kx))$$

$$\text{or } u^2 = \left(\frac{\omega}{k}\right)^2$$

$$\therefore u^2 = u^2$$

proved

Note :

$$\frac{d^2 y}{dt^2} = u^2 \cdot \frac{d^2 y}{dx^2}$$

Particle Acceleration = (Wave velocity)² × curvature of displacement

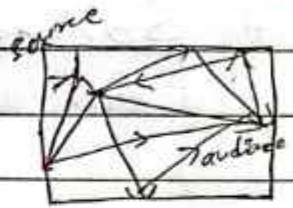
Unit - 3

classmate

Date _____
Page _____

Acoustics

Reverberation:



Standard Reverberation
time,

$$\frac{I_t}{I_0} = 10^{-6}$$

$$\Rightarrow I_t = 10^{-6} I_0$$

Absorption Coefficient
or

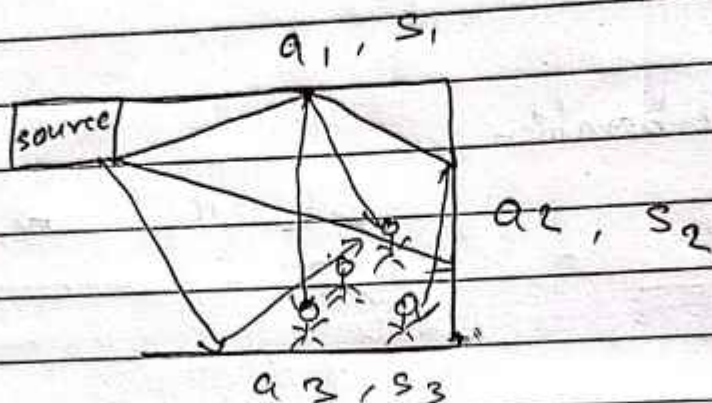
Absorption Power

[Open Window Unit] $\leftarrow$ It is Unit.
 $a = \text{OWU}$

$a = \frac{\text{amount of energy absorbed}}{\text{energy incident}}$

$$[\therefore \text{Reflecting power} = 1 - a]$$

Sabine's Formula [For Standard reverberation time]



$$\bar{a} = a_1 s_1 + a_2 s_2 + \dots$$

$$s_1 + s_2 + s_3 + \dots$$

$$\Rightarrow \bar{a} = \frac{\sum a s}{S} \quad \text{--- (1)}$$

$$\Rightarrow \sum a s = \bar{a} \cdot S \quad \text{--- (2)}$$

* Average distance travelled by a wave in a two successive reflection is $\frac{4V}{S}$ --- (3)

Let, v be the velocity of sound,
time betⁿ two successive reflection = $\frac{4V/S}{v}$

$$= \frac{4V}{Sv} \quad \text{--- (4)}$$

$V \rightarrow$ capital V

$v \rightarrow$ small v

(average $\rightarrow$ 1 reflection)

classmate
Date _____
Page _____

In $\frac{4V}{SV}$ time there is 1 reflection.

In 1 " " " " $\frac{SV}{4V}$ " "

In t " " " " $\left(\frac{SV \cdot t}{4V} \right)$ "

$\downarrow$

(no. of reflection on average)

$\bar{a}$ = average absorbing power
 $1 - \bar{a}$ = average reflecting power

• In two reflection, reflecting power = $(1 - \bar{a})^2$

In $\frac{SV \cdot t}{4V}$ reflection, reflecting power = $(1 - \bar{a})^{\frac{SV \cdot t}{4V}}$ — (5)

(3) For Standard Time,

$$I_t = 10^{-6} I_0$$

$$\Rightarrow \frac{I_t}{I_0} = 10^{-6} \quad \text{--- (6)}$$

From (5) & (6), we know,

$$10^{-6} = (1 - \bar{a})^{\frac{SV \cdot T}{4V}}$$

$$\text{or, } \log_e 10^{-6} = \log_e \left[(1 - \bar{a})^{\frac{SV \cdot T}{4V}} \right]$$

or, $2.3 \log_{10} 10^{-6} = \frac{SVT}{4V} \log_e (1-\bar{a})$

or, $(-6) \times 2.3 \log_{10} 10^{-6} = \frac{SVT}{4V} \left[-\bar{a} - \frac{\bar{a}^2}{2} - \frac{\bar{a}^3}{3} \dots \right]$

or $(-6) \times 2.3 = -\bar{a} \frac{SVT}{4V}$

or $T = \frac{13.8 \times 4V}{\bar{a} \cdot SV}$

or $T = \frac{55.2V}{\Sigma a_s \times 330}$

or $T = \frac{46V}{275 \Sigma a_s}$

$\left[\begin{aligned} \bar{a}s &= \Sigma a_s \\ V &= 330 \end{aligned} \right]$

∴ $T = \frac{0.167V}{\Sigma a_s}$

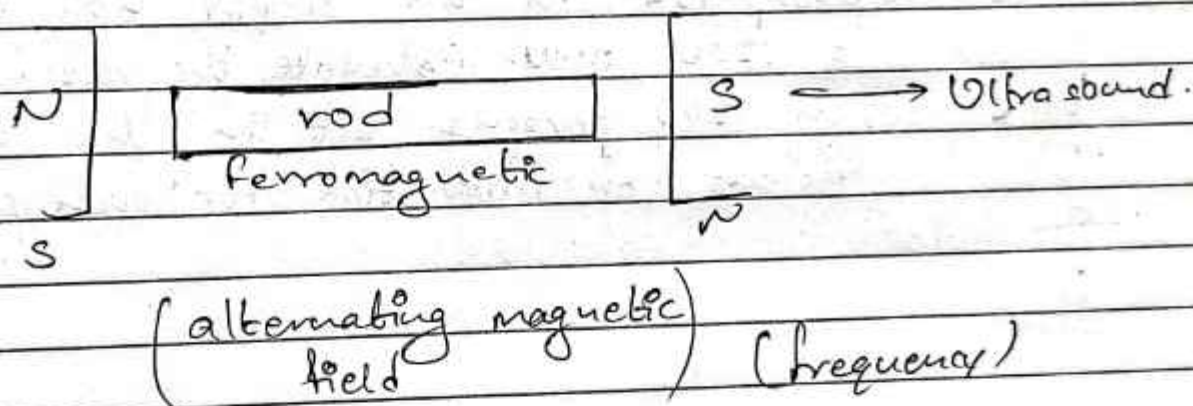
Ultra-Sound and its production.

Infra sound < 20 Hz to 20 kHz < Ultra sound
Audible range

1) Magnetostriction Method

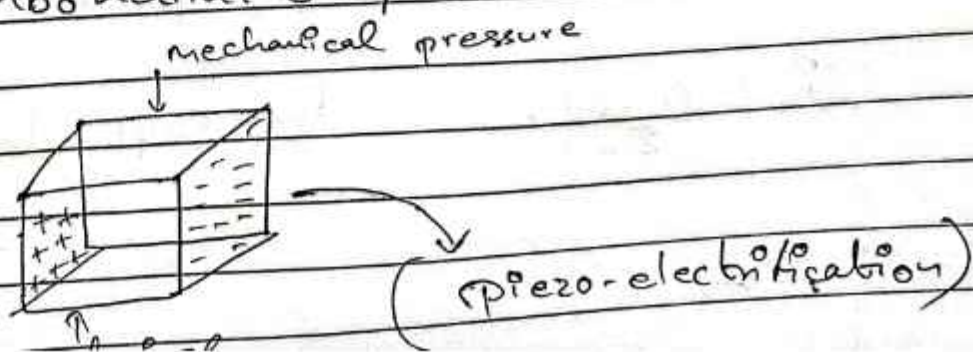
- strong ferromagnetic material.

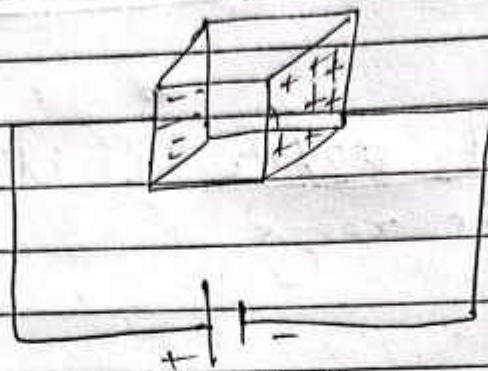
↓
(cobalt or nickel)



2) Piezo-electric method.

- Quartz Crystal.
(Rhomboidal Shaped)





(alternating potential)

→ Reverse piezo-electrification
(500 MHz frequency can be produced)
ultrasound

Numerical :-

- 1) Total absorption in an empty hall of volume 1200 m^3 is 120 osw. Calculate the reverberation time when 100 persons sit in that hall. The coefficient of absorption for each person is 0.7.

Soln

$$V = 1200 \text{ m}^3$$

$$\Sigma a_s = 120 \text{ osw}$$

$$T = ?$$

$$\text{no. of person} = 100$$

Coef. of absorption per person, $a_1 = 0.7$.

$$T = \frac{0.167V}{\Sigma a_s} \quad \text{— for empty hall}$$

Now,

For 100 people,

$$T_1 = \frac{0.167V}{\Sigma a_s + 100 \times a_1} = \frac{0.167 \times 1200}{120 + 100 \times 4.7} = 0.33 \text{ sec}$$

- 2) For an empty assembly hall of size $20 \times 15 \times 10 \text{ m}^3$, the reverberation time is 3.5 sec. Calculate the avg. absorption coeff of hall. What area of wall should be covered by curtain so as to reduce the reverberation time to 2.5 sec, the absorption coefficient of curtain is 0.5.

Soln

$$V = 20 \times 15 \times 10 \text{ m}^3$$

$$T_1 = 3.5 \text{ sec [For empty hall]}$$

$$\bar{a} = ? \text{ (avg. absorption coefficient)}$$

$$S_1 = ? \text{ (Area of curtain)}$$

$$T_2 = 2.5 \text{ sec (with curtain)}$$

Absorption coeff. of curtain, $a_1 = 0.5$

For empty hall,

$$T_1 = \frac{0.167V}{\Sigma a_s} \quad \text{--- (1)}$$

For Curtain,

$$T_2 = \frac{0.167V}{\Sigma a_s + a_1 S_1} \quad \text{--- (2)}$$

Form (1),

$$T_1 = \frac{0.167V}{\Sigma a_s}$$

$$\text{or } 3.5 = \frac{0.167 \times 20 \times 15 \times 10}{\Sigma a_s}$$

$$\therefore \Sigma a_s = 143.14$$

Now,

$$\begin{aligned} S &= 2(lb + bh + lh) \\ &= 2(20 \times 15 + 15 \times 10 + 10 \times 20) \\ &= 1300 \end{aligned}$$

$$\bar{a} = \frac{\Sigma a_s}{S} = \frac{143.14}{1300} = 0.11.$$

From (2),

$$T_2 = \frac{0.167 \times 20 \times 15 \times 10}{143.14 + 0.5 \times S_1}$$

$$\text{or } 2.5 = \frac{501}{143.14 + 0.55 S_1}$$

$$\text{or } 143.14 + 0.55 S_1 = \frac{501}{2.5}$$

$$\text{or } \frac{S_1}{2} = 57$$

$$\therefore S_1 = 114 \text{ m}^2$$

- 3) The reverberation time of an empty hall is 1.5 sec, with 500 audience, it becomes 1.4 sec then calculation of reverberation time with 800 audience in that hall.

Solⁿ

$$T_1 = \frac{0.167V}{\Sigma a_s} \text{ — for empty hall — (1)}$$

$$T_2 = \frac{0.167V}{\Sigma a_s + 800a_1} \text{ — for 800 audience — (2)}$$

$$T_2 = \frac{0.167V}{\Sigma a_s + 500a_1} \text{ — for 500 audience — (3)}$$

Now,

In (1),

$$T_1 = \frac{0.167V}{\Sigma a_s}$$

$$\text{as } 1.5 \times \Sigma a_s = 0.167V$$

$$\text{as } \Sigma a_s = 0.1113 V$$

In (2),

$$T_2 = \frac{0.167V}{\Sigma a_s + 500a_1}$$

$$\text{as } 1.4 = \frac{0.167V}{0.1113V + 500a_1}$$

$$\text{as } 0.1113V + 500a_1 = 0.1192V$$

$$\text{or, } 500a_1 = 0.0079V$$

$$\therefore a_1 = 1.58 \times 10^{-5} V$$

In (3),

$$\begin{aligned}
 T_3 &= 0.167 \text{ V} \\
 &\quad 0.1113 \text{ V} + 800 \times 1.58 \times 10^{-5} \text{ V} \\
 &= 0.167 \text{ V} \\
 &\quad 0.12394 \text{ V} \\
 &= 1.34 \text{ sec}
 \end{aligned}$$

Unit - 4

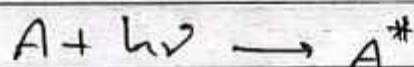
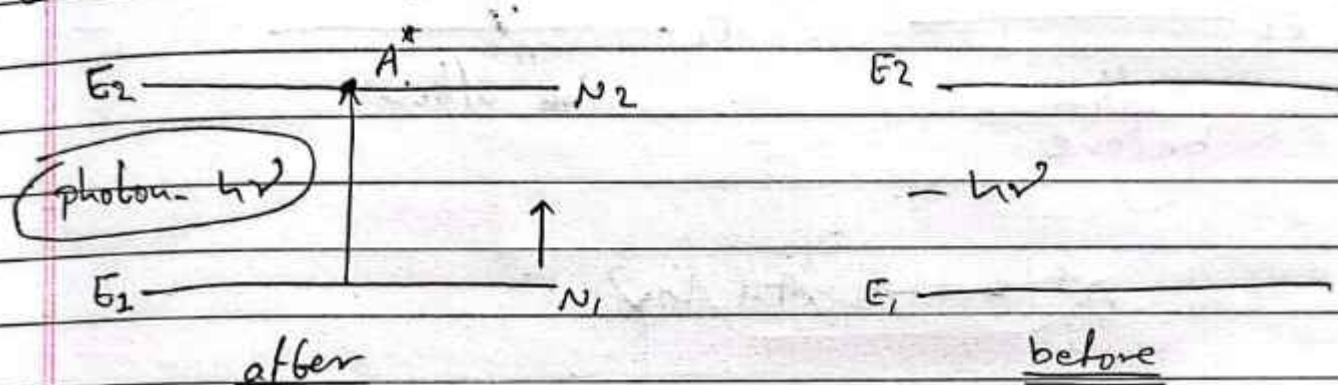
classmate

Date _____
Page _____

LASER & FIBER OPTICS

Laser - Light amplification by stimulated emission of radiation.

① Induced absorption



$\frac{dN_1}{dt}$ = rate of induced absorption, amount of atoms that makes upward transition.

$$\frac{dN_1}{dt} \propto N_2$$

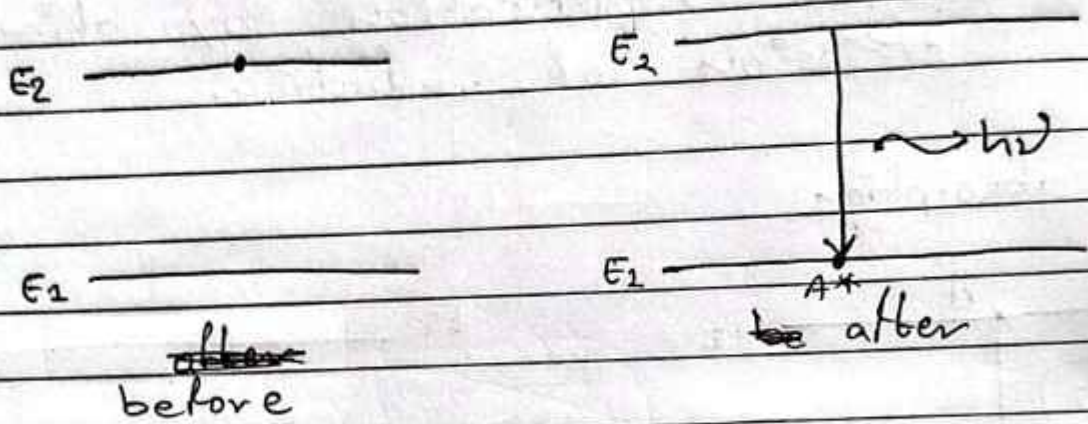
$$\frac{dN_1}{dt} \propto I_{\nu 1}$$

$$\frac{dN_1}{dt} \propto I_{\nu 1} N_2$$

$$\therefore -\frac{dN_1}{dt} = A_{21} I_{\nu 1} N_2$$

• Einstein's coeff. of induced absorption.

(II) Spontaneous emission



$$-\frac{dN_2}{dt} \propto N_2$$

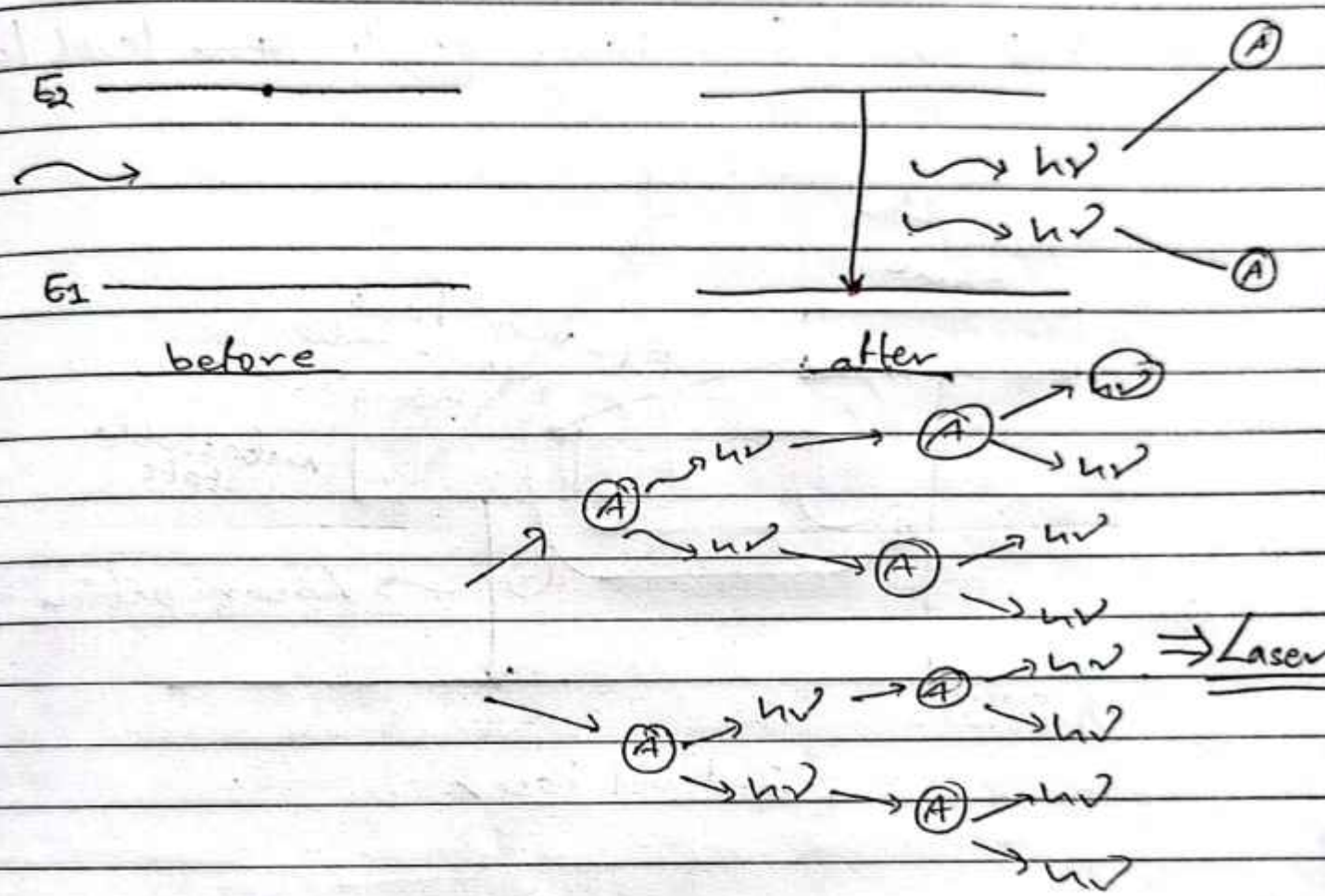
$$\Rightarrow -\frac{dN_2}{dt} = \beta_{21} N_2$$

$R_{spt.}$

Einstein coeff. of
spontaneous
emission.

→

III) Stimulated emission

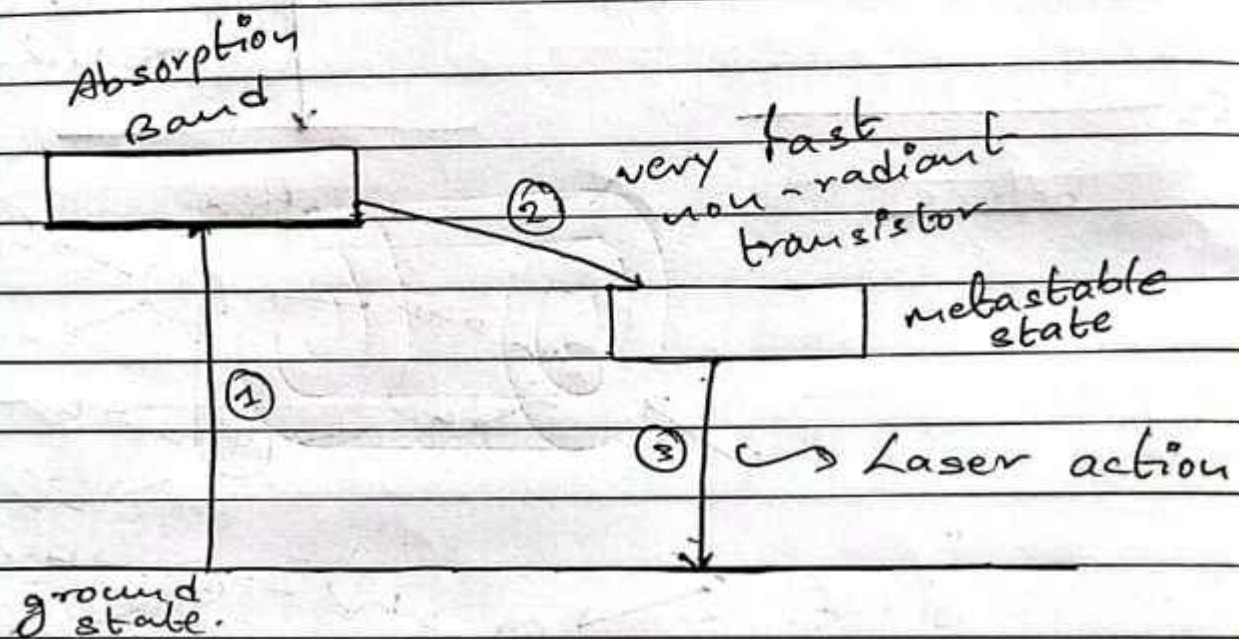


IV) Terms associated with Laser.

- 1) Population
- 2) Population inversion
- 3) Pumping
- 4) Meta-stable state
- 5) Active medium
- 6) Active center.

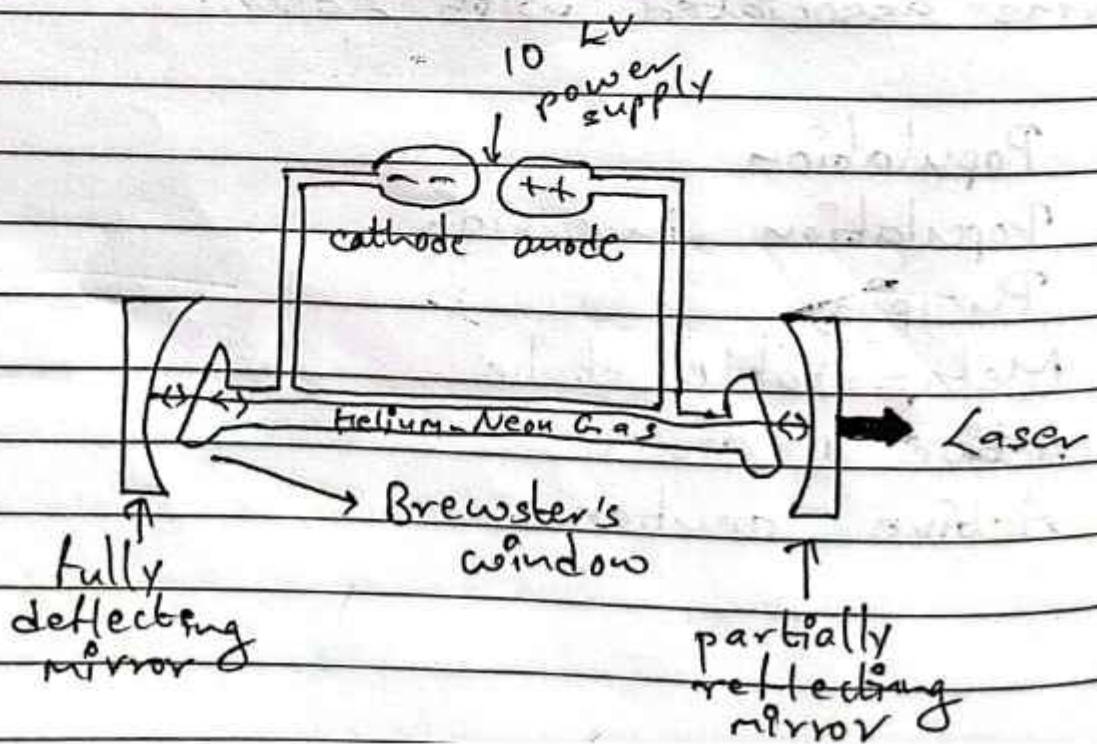
He-Ne Laser :-

(20:1) (Gas laser or 4-level laser)

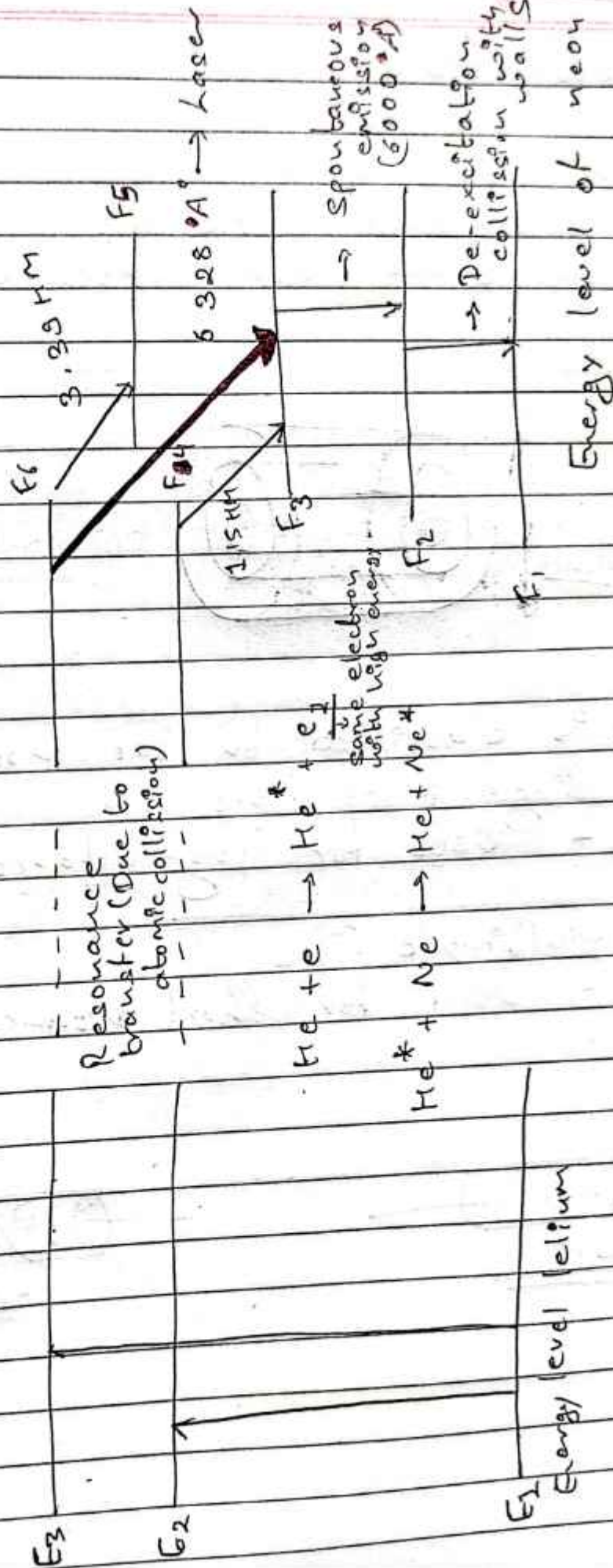


3-level laser

* Construction



a Working:-



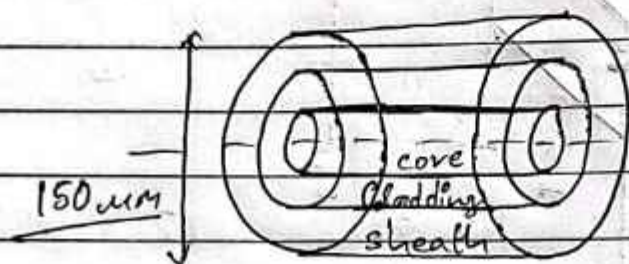
Population inversion betw
 F_6 & F_4 }
 F_5 & F_3 }
 F_4 & F_3 }

classmate

Date
Page

Optical fibre

- rod / pipe, it is.
- glass / plastic, made of.
- Principle → Total Internal Reflection.



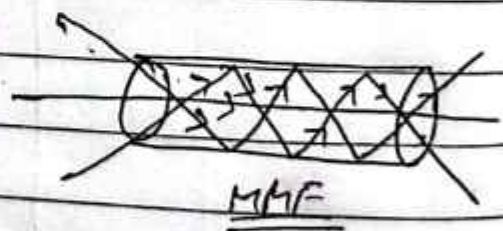
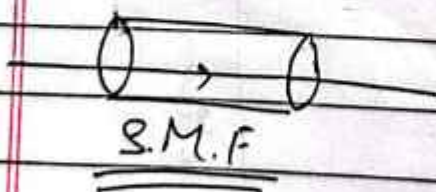
It is classified on the mode :-

1) Single Mode fibre

- Used for long-distance transmission

2) Multimode fibre

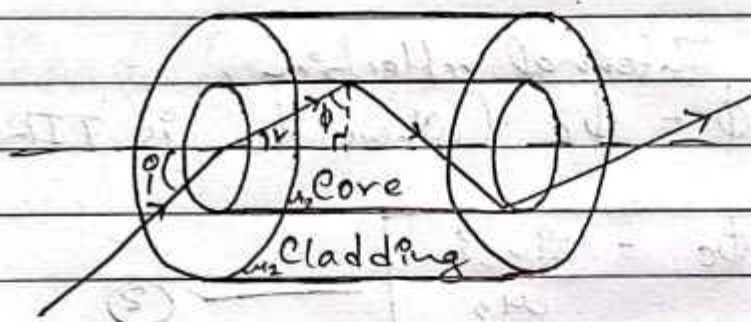
- Used for short-distance



SMF
┌──────────┐
Step Index Graded Index

MMF
┌──────────┐
Step Index Graded Index

* Acceptance Angle (θ)



• Snell's Law at incident edge,

$$\mu_2 \sin r = \mu_0 \sin i$$

$$\text{or, } \frac{\mu_2}{\mu_0} = \frac{\sin i}{\sin r}$$

$$\text{or } \mu_2 = \mu_0 \frac{\sin i}{\sin r}$$

$$\therefore \mu_2 = \frac{\sin i}{\sin r} \quad [\because \mu_0 = 1] \quad (\text{from air})$$

Now,

$$r + \phi = 90$$

$$\therefore r = 90 - \phi \quad \text{--- (1)}$$

Now, Taking $\sin$ in eqn (1),

$$\sin r = \sin(90 - \phi)$$

$$\sin r = \cos \phi$$

Here,

$$\mu_2 = \frac{\sin i}{\cos \phi}$$

$$\therefore \cos \phi = \frac{\sin i}{\mu_2}$$

For Total Internal reflection,

$\phi = \phi_c$ (where ϕ_c is TIR).

$$\therefore \cos \phi_c = \frac{\sin i}{\mu_2} \quad \text{--- (2)}$$

• Using Snell's law on Core-Cladding interface

$$\mu_1 \sin \phi_c = \mu_2 \sin 90^\circ$$

$$\text{or, } \mu_1 \sin \phi_c = \mu_2$$

$$\therefore \frac{\mu_1}{\mu_2} = \sin \phi_c$$

Now,

$$\sin^2 \theta = \sqrt{\mu_2^2 - \mu_1^2}$$

classmate

Date _____
Page _____

SG

Sq. Both sides,

$$\frac{\mu_1^2}{\mu_2^2} = \sin^2 \theta_c$$

$$\text{or, } \mu_1^2 = \sin^2 \theta_c \cdot \mu_2^2$$

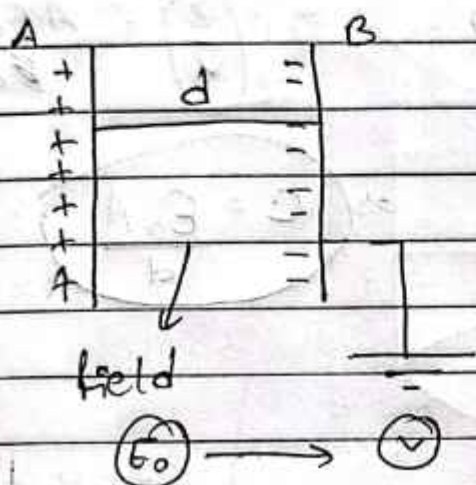
$$\text{or } \mu_1^2 = \sin^2 \theta_c$$

Normalized Frequency (ν)

Capacitor

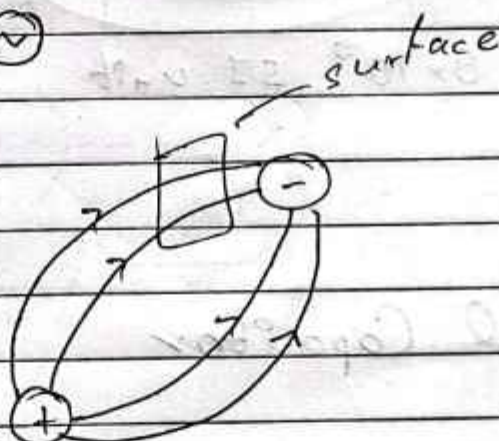
Types

1) Parallel Plate Capacitor



$$E = \frac{V}{d}$$

4



electric
flux, = Q_e

$$E = \frac{Q}{A}$$

$$Q_e = EA$$

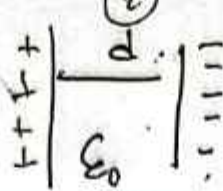
* Gauss Law of Electrostatics

$$Q_e = \frac{1}{\epsilon_0} \times \text{charge enclosed.}$$

ϵ_0 = permittivity in free space

ϵ = " in medium

$$\epsilon_r = \frac{\epsilon}{\epsilon_0}$$



$$Q_e = \frac{1}{\epsilon_0} q$$

$$E_0 = \frac{V}{d} \quad (2)$$

$$\Rightarrow \epsilon_0 A = \frac{q}{E_0}$$

$$\Rightarrow \frac{q}{A \epsilon_0} = \frac{V}{d}$$

$$\therefore \epsilon_0 = \frac{q}{A \cdot E_0} \quad (1)$$

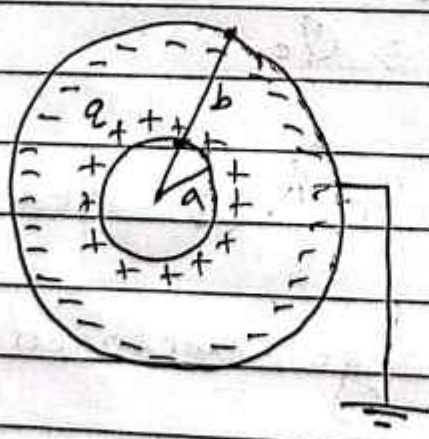
$$\Rightarrow \left(\frac{q}{V} \right) = \frac{A \epsilon_0}{d}$$

$$\Rightarrow C = \frac{\epsilon_0 A}{d}$$

- $\epsilon_0 = 8.85 \times 10^{-12} \text{ (S.I.)}$

- $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ S.I. unit}$

* Spherical Capacitor



$$E \rightarrow V \rightarrow C$$

$$V = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{\text{distance}}$$

$$V_A = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{a} \quad \text{--- (1)}$$

$$V_B = \frac{1}{4\pi\epsilon_0} \cdot \frac{(-q)}{b} \quad \text{--- (2)}$$

$$V = V_A + V_B$$

$$V = \frac{1}{4\pi\epsilon_0} \cdot q \cdot \left[\frac{1}{a} - \frac{1}{b} \right]$$

$$V = \frac{q}{4\pi\epsilon_0} \left[\frac{b-a}{ab} \right]$$

$$\frac{q}{V} = 4\pi\epsilon_0 \left(\frac{ab}{b-a} \right)$$

$$C = 4\pi\epsilon_0 \left(\frac{ab}{b-a} \right) \quad \text{--- (3)}$$

For single inner cell,

$$V = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{a}$$

$$\Rightarrow \frac{q}{V} = 4\pi\epsilon_0 a \quad \text{--- (4)}$$

$$\Rightarrow C' = 4\pi\epsilon_0 a \quad \text{--- (5)}$$

From (4) & (5),

$$\Rightarrow \frac{C}{C'} = \frac{4\pi\epsilon_0 \frac{ab}{b-a}}{4\pi\epsilon_0 a}$$

$$\Rightarrow \frac{C}{C'} = \frac{b}{b-a}$$

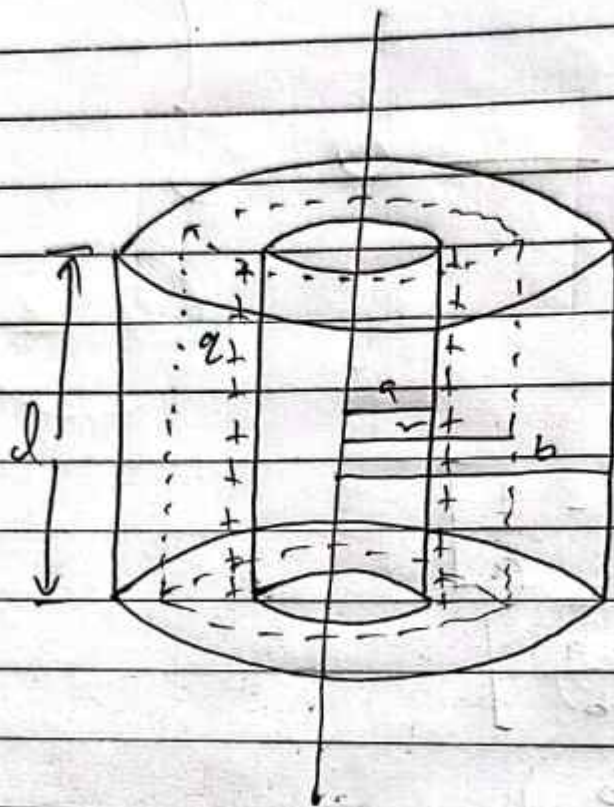
$$C = C' \cdot \frac{b}{b-a}$$

* Cylindrical Capacitor.

$$\Phi = \frac{1}{\epsilon_0} \times \text{charge enclosed}$$

$$E = \frac{V}{d \text{ i.e. } r \text{ for case}}$$

$$E = \frac{dv}{dr}$$



$$\Phi_e = \frac{1}{\epsilon_0} \times q$$

$$\Rightarrow EA = \frac{q}{\epsilon_0}$$

Area of
Gaussian
surface

$$\Rightarrow E = \frac{q}{A\epsilon_0} \quad \text{--- (1)}$$

$$E = - \frac{dv}{dr}$$

$$dv = - E dr$$

$$= - \frac{q}{A\epsilon_0} dr$$

$$= - q$$

$$\int dv = - \int_b^a \frac{q}{2\pi r l \epsilon_0} dr$$

$$V = - \frac{q}{2\pi l \epsilon_0} \left[\log r \right]_b^a$$

$$\Rightarrow V = - \frac{q}{2\pi l \epsilon_0} \left[\log_e a - \log_e b \right]$$

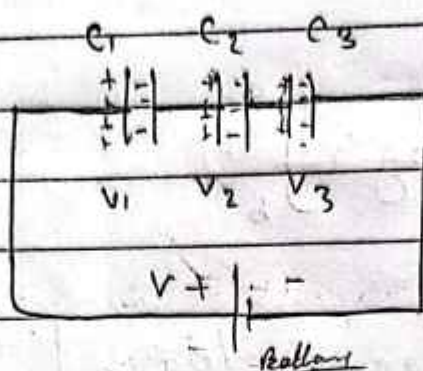
$$\Rightarrow V = - \frac{q}{2\pi l \epsilon_0} \left[\log_e \left(\frac{b}{a} \right) \right]$$

$$\Rightarrow \frac{q}{V} = \frac{2\pi l \epsilon_0}{\log_e (b/a)}$$

$$\therefore C = \frac{2\pi l \epsilon_0}{\log_e (b/a)}$$

Combination of Capacitors.

1) Series Combⁿ.



$$[V = q/c]$$

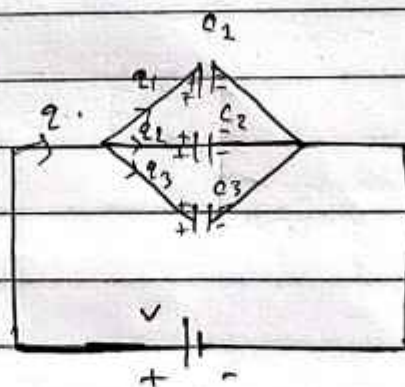
$$V = V_1 + V_2 + V_3$$

$$\text{or } \frac{q}{C} = \frac{q}{C_1} + \frac{q}{C_2} + \frac{q}{C_3}$$

$$\text{or } \frac{q}{C} = q \left[\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right]$$

$$\therefore \frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

2) Parallel Combⁿ.



$$[q = VC]$$

$$q = q_1 + q_2 + q_3$$

$$\text{or } V \cdot C = VC_1 + VC_2 + VC_3$$

$$\therefore C = C_1 + C_2 + C_3$$

Energy Stored in a Capacitor.

$$q = VC$$

$$\Rightarrow V = \frac{q}{C} \quad \text{--- (1)}$$

$dq \rightarrow dW = \text{potential energy} \times \text{charge}.$

$$dW = V \cdot dq.$$

$$\int dW = \int V dq$$

$$\Rightarrow W = \frac{1}{C} \cdot \frac{q^2}{2}$$

$$\left[\therefore \text{Energy} = \frac{1}{2} \frac{q^2}{C} \right] \rightarrow \text{Electrical energy.}$$

• Electrical energy density

$$U_e = \frac{\text{energy}}{\text{volume}}$$

$$\Rightarrow U_e = \frac{\frac{1}{2} \cdot \frac{q^2}{C}}{A \cdot d}$$

$$\Rightarrow U_e = \frac{1}{2} \frac{q^2}{C A d} \quad \text{--- (*)}$$

$$\therefore C = \frac{\epsilon_0 A}{d} \quad \text{--- (**)}$$

Magnetic Energy $\rightarrow (V_m = \frac{1}{2\mu_0} \dots)$

$$U_e = \frac{1}{2} \frac{dq^2}{Ad\epsilon_0 A}$$

$$\text{or } U_e = \frac{1}{2\epsilon_0 A^2} \cdot q^2 \quad (***)$$

According to Gauss Law,

$$Q_e = \frac{1}{\epsilon_0} \cdot q$$

$$\Rightarrow EA = \frac{q}{\epsilon_0}$$

(S.B.S)

$$\Rightarrow q^2 = E^2 A^2 \epsilon_0^2 \quad (****)$$

Here,

$$U_e = \frac{1}{2\epsilon_0 A^2} \cdot E^2 A^2 \epsilon_0^2$$

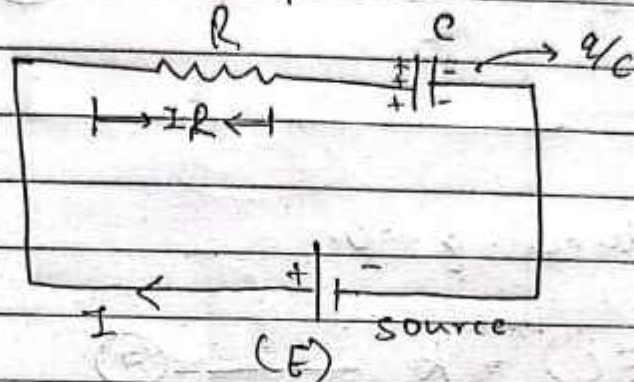
$$\therefore U_e = \frac{1}{2} \epsilon_0 \cdot E^2$$

#

Charging & Discharging of Capacitor through resistor.

(R-C circuit)

1) Charging of a Capacitor



C → capacitance

$$E = IR + \frac{q}{C} \quad \text{--- (1)}$$

Multiplying by 'C' in both side,

$$CE = IRC + q$$

$$\Rightarrow CE - q = \frac{dq}{dt} \cdot RC$$

$$\Rightarrow \frac{dq}{EC - q} = \frac{1}{RC} \cdot dt \quad \text{--- (2)}$$

$$\text{Let } EC - q = x,$$

$$\Rightarrow -\frac{dq}{dx} = 1 \quad \text{--- (3)}$$

$$\Rightarrow dq = -dx \quad \text{--- (4)}$$

From (3) & (4), we get in (2)

$$\int \frac{dq}{EC - q} = \int \frac{1}{RC} \cdot dt$$

$$\Rightarrow \int \frac{dx}{x} = - \int \frac{1}{RC} \cdot dt$$

$$\log_e x = -\frac{1}{RC} t + k \quad \text{--- (5)}$$

$$\Rightarrow \log_e (EC - q) = -\frac{t}{RC} + k \quad \text{--- (6)}$$

For k ,
 $t = 0$, then $q = 0$.

$$\Rightarrow \log_e (EC) = k \quad \text{--- (7)}$$

From (6) & (7),

$$\log_e (EC - q) = \frac{t}{RC} + \log_e (EC)$$

$$\Rightarrow \log_e \left[\frac{(EC - q)}{EC} \right] = -\frac{t}{RC}$$

$$\Rightarrow 1 - \frac{q}{EC} = e^{-t/RC}$$

$$\Rightarrow \frac{q}{EC} = 1 - e^{-t/RC}$$

$$\Rightarrow q = EC [1 - e^{-t/RC}] \quad \text{--- (8)}$$

When $t = 0$, $q = 0$

$t = \infty$, $q = EC \rightarrow$ maximum charge.
 $q = q_0$.

$$\therefore q = q_0 (1 - e^{-t/RC}) \quad \text{--- (9)}$$

$$\left[\begin{array}{l} \frac{t}{RC} = \text{constant} \\ \quad \quad \quad (\text{time constant}) \\ RC = \text{dimension of time} \\ \frac{t}{RC} = 1 \rightarrow 1 = \text{time constant} \end{array} \right]$$

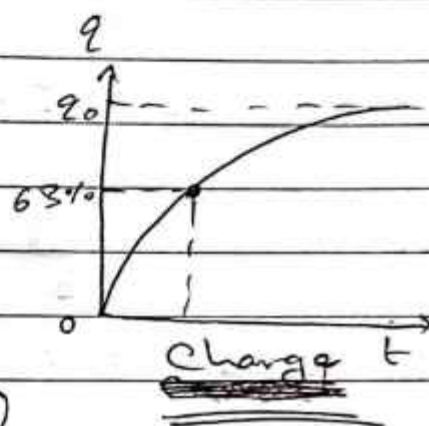
At 1 - time constant,

$$q = q_0 [1 - e^{-1}]$$

$$\Rightarrow q = q_0 [1 - 0.37]$$

$$\Rightarrow q = 0.63 q_0$$

$$\therefore q = 63\% \text{ of } q_0 \quad \text{--- (i)}$$



• Current at time 't' = ?

$$\therefore I = \frac{dq}{dt} = \frac{d}{dt} q_0 [1 - e^{-t/RC}]$$

$$= \frac{q_0 \cdot e^{-t/RC}}{RC}$$

$$= \frac{q_0/c}{R} e^{-t/RC}$$

$$[q_0/c = E]$$

$$= \frac{E}{R} e^{-t/RC}$$

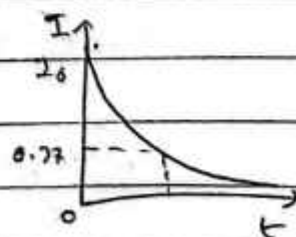
$$[E/R = I_0]$$

$$\therefore I = I_0 e^{-t/RC}$$

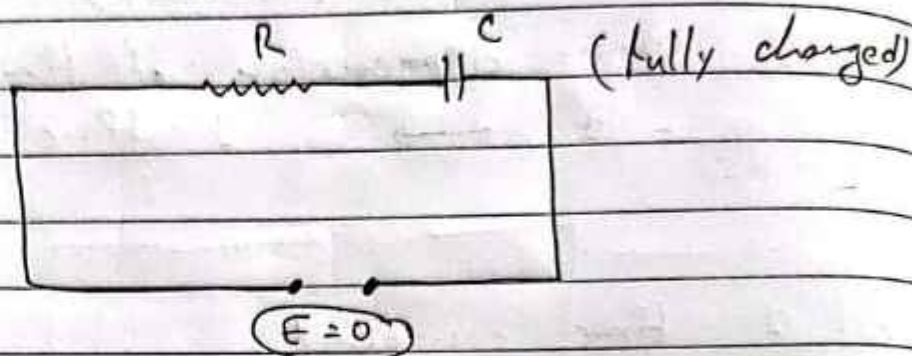
At 1-time constant, $t = RC$, then

$$I = I_0 \cdot e^{-1}$$

$$\therefore I = 0.37 I_0 \quad \text{--- (ii)}$$



2) Discharging of Capacitor



$$0 = IR + \frac{q}{C}$$

$$\therefore 0 = \frac{dq}{dt} \cdot R + \frac{q}{C}$$

$$\therefore \frac{dq}{dt} \cdot R = -\frac{q}{C}$$

$$\therefore \frac{dq}{q} = -\frac{1}{RC} \cdot dt \quad \text{--- (1)}$$

$$\int \frac{dq}{q} = \int -\frac{1}{RC} \cdot dt$$

$$\therefore \log_e q = -\frac{1}{RC} t + k \quad \text{--- (2)}$$

When $t = 0$, $q = q_0$ (or EC)

$$\therefore \log_e q_0 = k \quad \text{--- (3)}$$

From (2) & (3),

$$\log_e q = -\frac{1}{RC} t + \log_e q_0$$

$$\text{or, } \log_e \left(\frac{q}{q_0} \right) = -\frac{t}{RC}$$

$$\text{or } q = q_0 \cdot e^{-t/RC} \quad \text{--- (14)}$$

When, $t = 0, q = q_0,$
 $t = \infty, q = 0.$

At 1-time constant,

$$\frac{t}{RC} = 1$$

$$\therefore q = q_0 e^{-1}$$

$$\text{or } q = \frac{q_0}{e}$$

$$\therefore q = 0.37 q_0$$

$$\therefore q = 37\% \text{ of } q_0.$$

• Current,

$$I = \frac{dq}{dt}$$

$$\frac{dq}{dt} = q_0 \left(\frac{-1}{RC} \right) e^{-t/RC}$$

$$\text{or } I = -\frac{q_0}{RC} \cdot e^{-t/RC}$$

$$\text{or } I = -I_0 \cdot e^{-t/RC}$$

$$\left[\frac{q_0}{C} = \frac{V}{R} = I_0 \right]$$

Here, $I = -0.37 I_0$

Dielectrics

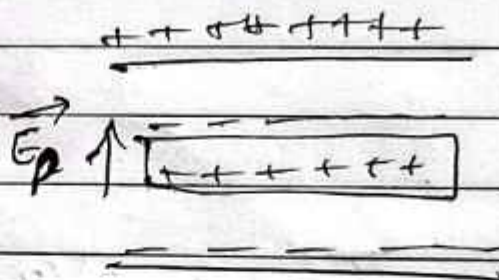
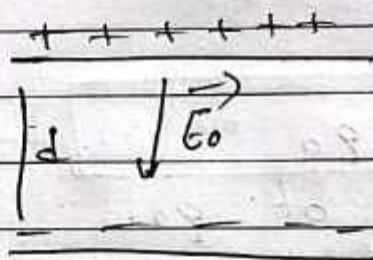
Polar
Dielectric

Non-Polar
Dielectric

• Dielectric constants

$$k = \frac{\epsilon_0}{\epsilon} = \frac{\epsilon_0}{\epsilon_0 - \epsilon_p}$$

Since, $k > 1$.



• Polarization,

$$P = \frac{\text{dipole moment}}{\text{volume}} \quad (p)$$

$$= \frac{q'l}{A \times l}$$

$$\therefore \vec{P} = \frac{q'}{A} \quad \text{--- (4)}$$

$\vec{E}$ = electric vector
 $\vec{D}$ = Displacement vector
 $\vec{P}$ = Polarization vector.

* Relation betⁿ $\vec{E}$, $\vec{D}$ and $\vec{P}$.

+++++

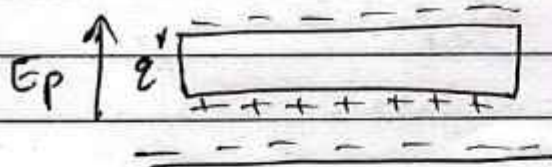
$\downarrow E_0$ (2) $\rightarrow$

$$Q_e = \frac{1}{\epsilon_0} \times q$$

$$\Rightarrow E_0 \cdot A = \frac{q}{\epsilon_0}$$

$$\Rightarrow E_0 = \frac{q}{A\epsilon_0} \quad \text{--- (1)}$$

+++++



$$Q_e = \frac{1}{\epsilon_0} (q - q')$$

$$\text{or, } (E_0 - E_p)A = \frac{q - q'}{\epsilon_0}$$

$$\text{or } E = \frac{q - q'}{A\epsilon_0} \quad \text{--- (2)}$$

We have,

$$k = \frac{E_0}{E}$$

$$\text{or } k = \frac{q/A\epsilon_0}{(q - q')/A\epsilon_0}$$

$$\text{or } k = \frac{q}{q - q'}$$

$$\text{or } \frac{1}{k} = \frac{q - q'}{q}$$

$$\text{or, } \frac{1}{k} = 1 - \frac{q'}{q}$$

$$\text{or, } \frac{q'}{q} = 1 - \frac{1}{k}$$

$$\text{or } q' = q \left[1 - \frac{1}{k} \right]$$

$$\downarrow$$

$$k > 1$$

$$q' < q$$

$$\text{If } k = 1 \rightarrow q = q' = 0.$$

From (2),

$$E = \frac{q - q'}{A \epsilon_0}$$

$$\text{or } E = \frac{q}{A \epsilon_0} - \frac{q'}{A \epsilon_0}$$

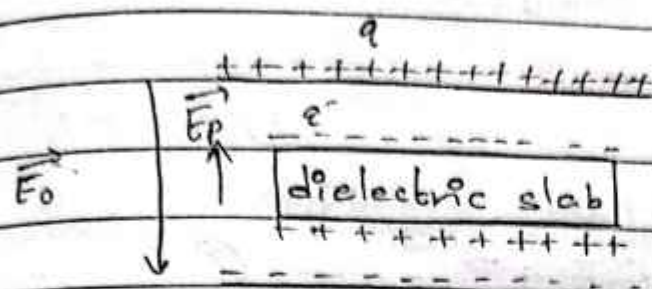
$$\text{or } \epsilon_0 E = \frac{q}{A} - \frac{q'}{A}$$

$$\text{or } \epsilon_0 E = D - P$$

$$\therefore D = \epsilon_0 E + P$$

##

Clausius Mosotti Equation



$$\vec{E} = \vec{E}_0 - \vec{E}_p \quad \text{Resultant field.} \quad \text{--- (1)}$$

$\vec{p}$: polarization vector

P = displacement
volume

$$P = \frac{q \cdot d}{A \cdot d}$$

$$\therefore P = \frac{q'}{A} \quad \text{--- (2)}$$

Internal force experienced by molecules = E_i

$$\therefore E_i = E + \frac{P}{3\epsilon_0} \quad \text{--- (3)}$$

$\underbrace{\quad}_{\text{External field}} \quad \underbrace{\quad}_{\text{Polarizing field}}$

Here,

$$P \propto E_i$$

$$\Rightarrow P = \alpha E_i \quad \text{--- (4)}$$

$\downarrow$
 coefficient of
 polarization
 or
 polarizability.

$$P = \alpha \left[E + \frac{P}{3\epsilon_0} \right] \text{-----} (5)$$

Also,

We have, $D = \epsilon E \text{-----} (6)$

Also, we have, $D = \epsilon_0 E + P \text{-----} (7)$

$$\Rightarrow \epsilon E = \epsilon_0 E + P$$

$$\Rightarrow E(\epsilon - \epsilon_0) = P$$

$$\Rightarrow E = \frac{P}{\epsilon - \epsilon_0} \text{-----} (8)$$

From (5) & (8),

$$P = \alpha \left[\frac{P}{\epsilon - \epsilon_0} + \frac{P}{3\epsilon_0} \right]$$

$$\Rightarrow 1 = \alpha \left[\frac{1}{\epsilon - \epsilon_0} + \frac{1}{3\epsilon_0} \right]$$

If there are N . no. of molecules,

$$1 = N\alpha \left[\frac{1}{\epsilon - \epsilon_0} + \frac{1}{3\epsilon_0} \right]$$

$$\Rightarrow 1 = N\alpha \left(\frac{3\epsilon_0 + \epsilon - \epsilon_0}{3\epsilon_0(\epsilon - \epsilon_0)} \right)$$

$$\Rightarrow 1 = \frac{N\alpha}{3\epsilon_0} \left(\frac{2\epsilon_0 + \epsilon}{\epsilon - \epsilon_0} \right) \text{-----} (9)$$

Also, $\epsilon_r = \frac{\epsilon}{\epsilon_0}$

$$\Rightarrow \epsilon = \epsilon_r \cdot \epsilon_0$$

(10)

$\text{Ions} \Rightarrow \alpha = \alpha_i$
 $\text{Electronic} \Rightarrow \alpha = \alpha_e$

classmate

Date

Page

$$1 = \frac{N\alpha}{3E_0} \left[\frac{2E_0 + E_r \cdot E_0}{E_r \cdot E_0 - E_0} \right]$$

$$\therefore 1 = \frac{N\alpha}{3E_0} \left[\frac{E_r + 2}{E_r - 1} \right]$$

↓

Relation betⁿ E_r and α .

Numerical

Q. What distance should the plates each of area $0.2 \text{ m} \times 0.1 \text{ m}$ of an air capacitor be placed in order to have same capacitance of spherical capacitor of radius 0.5 m .

Solⁿ

$$\text{Spherical capacitor } (C) = 4\pi\epsilon_0 r$$

$$\text{Parallel " } (C') = \frac{\epsilon_0 A}{d}$$

Now,

We know the condition of same capacitance,
 $C = C'$

$$\text{or } 4\pi\epsilon_0 r = \frac{\epsilon_0 A}{d}$$

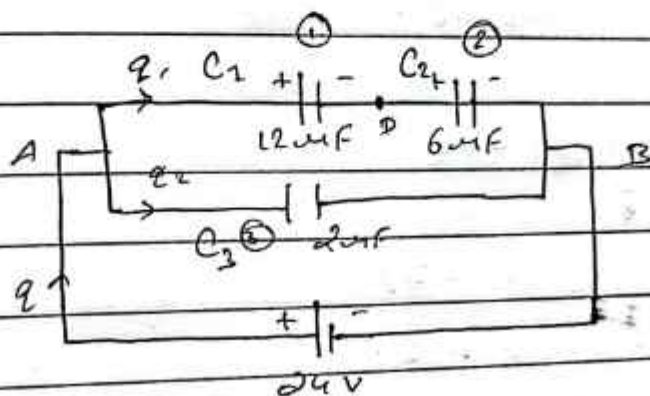
$$\text{or } 4\pi r \cdot d = A$$

$$\text{or } d = \frac{A}{4\pi r}$$

$$\text{or } d = \frac{0.2 \times 0.1}{4 \times \pi \times 0.5}$$

$$\therefore d = 3.18 \times 10^{-3} \text{ m}$$

- Q) Consider the circuit as shown in fig.
 Find. i) total capacitance of the circuit.
 ii) charges on each capacitor.
 iii) potential at 'D' assuming zero potential drop at 'B'.



Solⁿ

$$V = 24 \text{ V}$$

C_1 & C_2 are in series and they are in parallel with C_3 .

$$\frac{1}{C'} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{12} + \frac{1}{6}$$

$$\therefore C' = 4 \mu\text{F}$$

Here,

$$\begin{aligned} C &= C' + C_3 \\ &= 4 + 2 \\ &= 6 \mu\text{F} \end{aligned}$$

Now,

$$\begin{aligned} \text{Total charge } (q) &= VC \\ &= 24 \times 6 \\ &= 144 \mu\text{C} \end{aligned}$$

We know,

$$V = V_3 = V_1 + V_2 = 24 \text{ V}$$

$$\begin{aligned} q_2 &= V \times C_3 \\ &= 24 \times 2 \\ &= 48 \mu\text{C} \end{aligned}$$

Now,

$$\begin{aligned} q &= q_1 + q_2 \\ \text{or } q - q_2 &= q_1 \\ \text{or } q_1 &= 144 - 48 \\ \therefore q_1 &= 96 \mu\text{C} \end{aligned}$$

$\therefore$ Potential at C_2 , $(D = C_2)$.

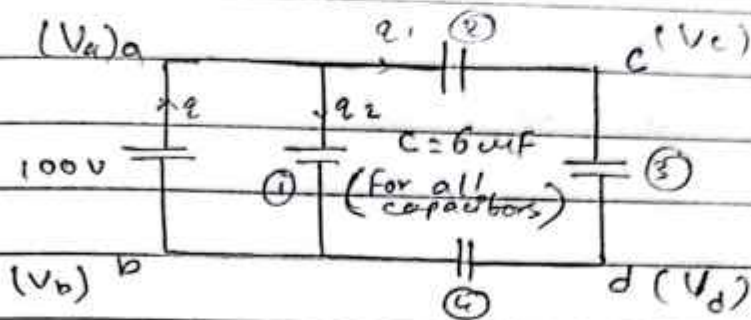
$$\frac{q_1}{C_2} = V_2$$

$$\text{or } V_2 = \frac{96}{6}$$

$$\therefore V_2 = 16 \text{ V}$$

potential = V

Q) Consider a circuit as shown in fig. Assuming zero potential drop at 'b' find the potential betⁿ 'c' and 'd'.



Solⁿ

$$V_c - V_d = ?$$

$$V_b = 0$$

$$V_a = 100 \text{ V} \quad \text{--- (1)}$$

Charge in 2nd capacitor,

$$q_1 = (V_a - V_c) \cdot 6$$

$$q_2 = (100 - V_c) \cdot 6 \quad \text{--- (2)}$$

Charge on 3rd capacitor,

$$q_3 = (V_c - V_d) \cdot 6 \quad \text{--- (3)}$$

Charge on 4th capacitor,

$$q_4 = V_d \cdot 6 \quad \text{--- (4)}$$

Now,

From (2) & (3),

$$(100 - V_c) \cdot 6 = (V_c - V_d) \cdot 6$$

$$\Rightarrow 100 - V_c = V_c - V_d$$

$$\Rightarrow 100 = 2V_c - V_d$$

$$\therefore V_c = \frac{100 + V_d}{2} \quad \text{--- (5)}$$

From ③ & ④,

$$(V_c - V_d) \cdot 6 = V_d \cdot 6$$

$$\Rightarrow V_c - V_d = V_d$$

$$\Rightarrow V_c = 2V_d$$

$$\therefore V_c = 2V_d \quad \text{--- ⑥}$$

From ⑤ & ⑥,

$$\frac{100 + V_d}{2} = 2V_d$$

$$\Rightarrow 100 + V_d = 4V_d$$

$$\Rightarrow 100 = 3V_d$$

$$\therefore V_d = 33.33$$

$$\therefore V_c = 66.67$$

Here,

$$q_1 = V_d \cdot 6$$

$$= 33.33 \times 6$$

$$= 200 \text{ nC.}$$

Potential betⁿ C and d,

$$V_c - V_d = 66.67 - 33.33$$

$$= 33.34$$

P
P
P
P
P

Q. A parallel plate capacitor has a capacitance of $100 \times 10^{-12} \text{ F}$, a plate of area of 100 cm^2 mica is used as a dielectric with dielectric constant 7. At 50 volts p.d. calculate the electric field intensity and magnitude of induced charge.

Solⁿ

$$C = 100 \times 10^{-12} \text{ F}$$

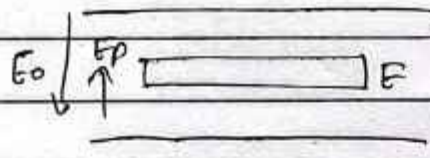
$$A = 100 \text{ cm}^2 = 0.01 \text{ m}^2$$

$$K = 7$$

$$V = 50 \text{ V.}$$

$$E = ? \quad (\text{Electric field intensity})$$

$$q' = ? \quad (\text{magnitude of induced charge}).$$



$$K = \frac{E_0}{E} \quad \text{--- (1)}$$

According to Gauss Law,
Before introducing slab,

$$Q = \frac{1}{\epsilon_0} \times q$$

$$\Rightarrow E_0 \cdot A = \frac{q}{\epsilon_0}$$

$$\text{or } E_0 = \frac{q}{A \epsilon_0} \quad \text{--- (2)}$$

Now,

After introducing slab,

$$Q = \frac{1}{\epsilon_0} (q - q')$$

$$\therefore EA = \frac{q - q'}{\epsilon_0}$$

$$\therefore E = \frac{1}{A \epsilon_0} (q - q') \quad \text{--- (3)}$$

In eqⁿ (3),

$$k = \frac{E_0}{E}$$

$$\therefore \gamma = \frac{q / A \epsilon_0}{(q - q') / A \epsilon_0}$$

$$\therefore \gamma = \frac{q}{q - q'}$$

$$\therefore \gamma q - \gamma q' = q$$

$$\therefore 6q = \gamma q'$$

$$\therefore q' = \frac{6q}{\gamma}$$

$$\therefore q' = \frac{6 \times 50 \times 100 \times 10^{-12}}{\gamma}$$

$$\therefore q' = \frac{6 \times 50 \times 100 \times 10^{-12}}{\gamma}$$

$$\therefore q' = 4.285 \times 10^{-9}$$

Here,

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ SI unit}$$

2.22 x 10⁻¹¹



P
P
P
P
P

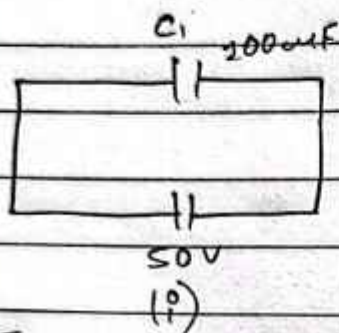
$$E = \frac{1}{A\epsilon_0} (q - q')$$

$$= \frac{1}{0.01 \times 8.85 \times 10^{-12}} (V.C - q')$$

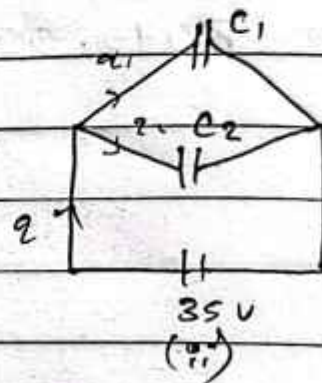
$$= \frac{1}{0.01 \times 8.85 \times 10^{-12}} (50 \times 100 \times 10^{-12} - 4.285 \times 10^{-9})$$

$$= \cancel{8079} \times 8079.09$$

Q) A $100 \mu\text{F}$ capacitor is charged to a potential difference of 50 volts . The charging battery then being disconnected, the capacitor is then connected to 2nd capacitor in parallel. If the measured potential drop of 35 V what is the capacitance of 2nd capacitor?



$C_2 = ?$



Soln

$$C_1 = 100 \mu\text{F}$$

In (i),

$$\begin{aligned} q &= VC \\ &= 50 \times 100 \times 10^{-6} \\ &= 5 \times 10^{-3} \quad \text{--- (1)} \end{aligned}$$

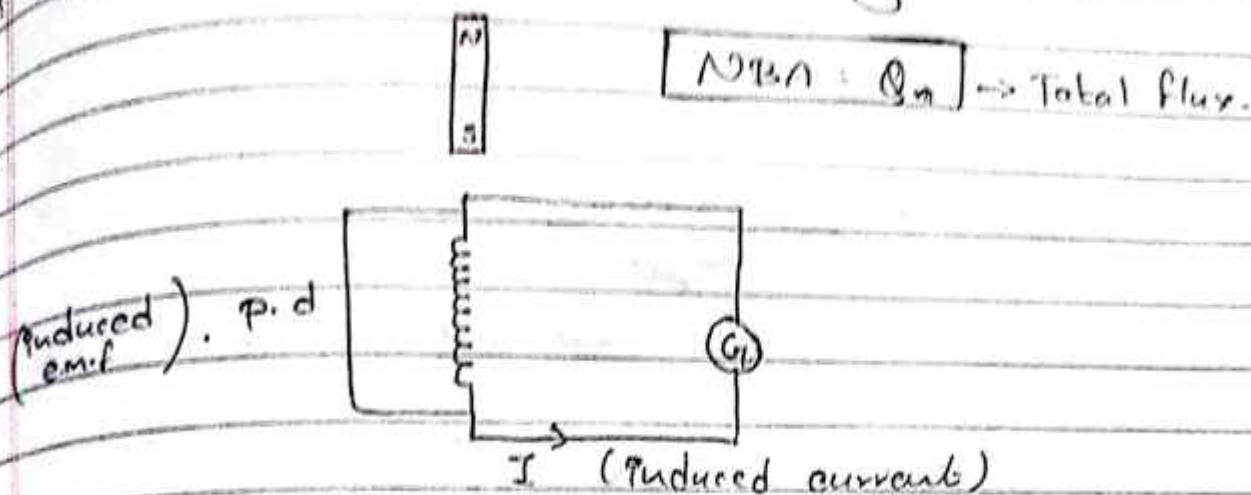
In (ii), $V' = 35 \text{ V}$

$$\begin{aligned} q &= q_1 + q_2 \\ \therefore q &= V'C_1 + V'C_2 \\ \therefore q &= V'(C_1 + C_2) \\ \therefore \frac{5 \times 10^{-3}}{35} &= 100 + C_2 \end{aligned}$$

$$\begin{aligned} \text{or, } 1.42 \times 10^{-4} &= 100 \times 10^{-6} + C_2 \\ \therefore C_2 &= (1.42 \times 10^{-4}) - (100 \times 10^{-6}) \\ \therefore C_2 &= 4.28 \times 10^{-5} \end{aligned}$$

Electro Magnetism

* Faraday's Law of Electromagnetic Induction.

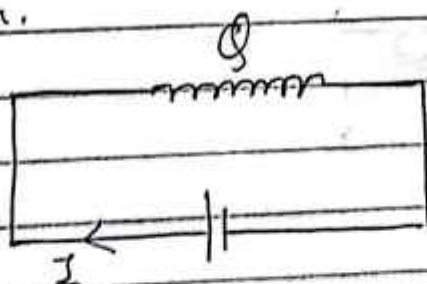


$$e \propto \frac{d\Phi_m}{dt}$$

$$\Rightarrow e = -k \frac{d\Phi_m}{dt}$$

$$\Rightarrow e = - \frac{d\Phi_m}{dt} \quad \text{--- (1)}$$

* Self Induction.



$$\Phi \propto I$$

$$\Rightarrow \Phi = L I$$

↓
(coeff. of self. inductance)

$$\Rightarrow L = \frac{\Phi}{I} \quad \text{--- (2)}$$

$$I = \frac{d\Phi}{dt}$$

$$\Rightarrow \boxed{d\Phi = I \cdot dt}$$

classmate
Date _____
Page _____

* Energy Stored in Inductor
(Work done against back emf).

$$\text{potential} = \frac{\text{work done}}{\text{charge}}$$

$$\Rightarrow \text{work done} = \text{potential} \times \text{charge}$$

$$\Rightarrow dW = L \frac{dI}{dt} \times I dt$$

$$\Rightarrow \int dW = \int L I \cdot dI$$

$$W = \frac{1}{2} L I^2$$

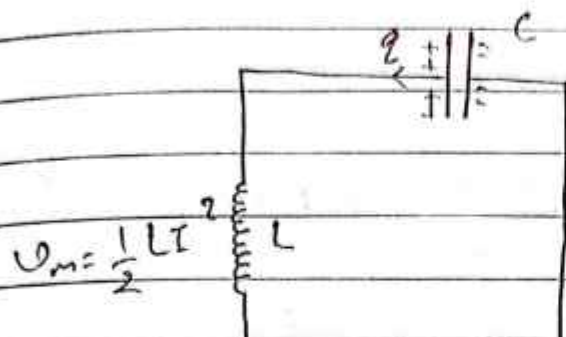
↓
magnetic energy,

$$U_m = \frac{1}{2} L I^2$$

$$U_e = \frac{1}{2} \frac{q^2}{C}$$

↓
electrical
energy,

Free Oscillation (L-C Oscillation):



$$U_e = \frac{1}{2} \frac{q^2}{C}$$

$$U_m = \frac{1}{2} L I^2$$

$$U = U_e + U_m$$

$$\Rightarrow U = \frac{1}{2} \frac{q^2}{C} + \frac{1}{2} L I^2$$

$$\Rightarrow \frac{dU}{dt} = \frac{1}{2} \frac{2q}{C} \left(\frac{dq}{dt} \right) + \frac{1}{2} \cdot L \cdot 2I \cdot \frac{dI}{dt}$$

$$\Rightarrow 0 = \frac{q}{C} \cdot I + L I \frac{d^2 q}{dt^2}$$

$$\Rightarrow L \frac{d^2 q}{dt^2} + \frac{q}{C} = 0$$

$$\Rightarrow \frac{d^2 q}{dt^2} + \frac{1}{LC} q = 0 \quad \text{--- (*)}$$

$$I = \frac{dq}{dt}$$

$$\frac{dI}{dt} = \frac{d^2 q}{dt^2}$$

Standard form of free oscillation,

$$\frac{d^2 x}{dt^2} + \omega^2 x = 0 \quad \text{--- (**)}$$

$$\therefore \omega^2 = \frac{1}{LC}$$

$$\Rightarrow \omega = \frac{1}{\sqrt{LC}}$$

$$2\pi f_0 \rightarrow \left[f_0 = \frac{1}{2\pi\sqrt{LC}} \right] \rightarrow \text{natural resonant.}$$

Alternative method

from (14) $\frac{d^2x}{dt^2} = -\omega^2 x$ — (1)

Let, $x = A \sin(\omega t + \delta)$ — (2)

$\frac{d^2x}{dt^2} = -\omega^2 A \sin(\omega t + \delta)$ — (3)

put $\rightarrow \boxed{x \rightarrow \delta}$

finally substitute in eqⁿ (1),

$$-\omega^2 A \sin(\omega t + \delta) + \frac{1}{LC} A \sin(\omega t + \delta) = 0$$

$$\text{or } \frac{1}{LC} A \sin(\omega t + \delta) = \omega^2 A \sin(\omega t + \delta)$$

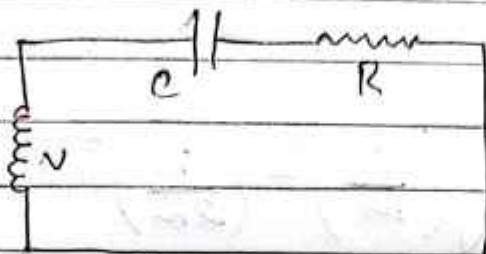
$$\text{or } \frac{1}{LC} = \omega^2$$

$$\text{or } \omega = \frac{1}{\sqrt{LC}}$$

$$\therefore T = \frac{1}{2\pi\sqrt{LC}}$$

$U_e \rightarrow$ Electrical energy
 $U_m \rightarrow$ Magnetic energy.

Damped L-C Oscillation:



$$U = U_e + U_m$$

$$\Rightarrow U = \frac{1}{2} L I^2 + \frac{1}{2} \frac{q^2}{C}$$

$$\Rightarrow \frac{dU}{dt} = \cancel{2I} \cdot L \frac{dI}{dt} + \frac{1}{\cancel{2}} \frac{2q}{C} \cdot \frac{dq}{dt}$$

$$\Rightarrow \frac{dU}{dt} = L I \frac{dI}{dt} + \frac{q}{C} \frac{dq}{dt}$$

$$\Rightarrow -I^2 R = L I \frac{d^2 q}{dt^2} + \frac{q}{C} I$$

$$\Rightarrow -\frac{dq}{dt} \cdot R = L \frac{d^2 q}{dt^2} + \frac{q}{C}$$

$$\Rightarrow L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C} = 0$$

$$\therefore \frac{d^2 q}{dt^2} + \frac{R}{L} \frac{dq}{dt} + \frac{1}{LC} q = 0 \quad \text{--- (1)}$$

Now,

We have,

Standard eqⁿ of damped oscillation,

$$\frac{d^2 x}{dt^2} + \frac{b}{m} \frac{dx}{dt} + \frac{k}{m} x = 0 \quad \text{--- (2)}$$

soln of ②,

$$x = A e^{-\frac{bt}{2m}} \sin(\omega' t + \delta) \quad \text{--- ③}$$

We know,

$$\omega' = \sqrt{\left(\frac{K}{m}\right) - \left(\frac{b}{2m}\right)^2}$$

Then,

$$\frac{K}{m} = \frac{1}{LC}$$

$$\frac{b}{2m} = \frac{R}{L}$$

$$\Rightarrow \omega' = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$$

$$\Rightarrow 2\pi f = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$$

$$\Rightarrow f = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$$

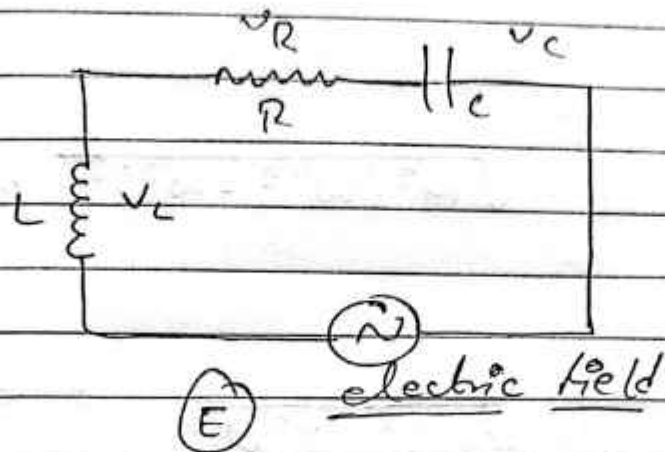
damped frequency.

$$a_1 \quad \frac{1}{LC} = \frac{R^2}{4L^2}$$

$$a_1 \quad \frac{1}{LC} < \frac{R^2}{4L^2}$$

$$a_1 \quad \frac{1}{LC} > \frac{R^2}{4L^2}$$

Forced Oscillation (L-C-R oscillation).



$$E = E_0 \sin \omega t \quad \text{--- (1)}$$

$$E = V_L + V_R + V_C$$

$$\Rightarrow E_0 \sin \omega t = L \frac{dI}{dt} + IR + \frac{q}{C}$$

$$\Rightarrow \frac{dI}{dt} + \frac{R}{L} I + \frac{1}{LC} q = E_0 \sin \omega t$$

$$\Rightarrow \frac{d^2 q}{dt^2} + \frac{R}{L} \frac{dq}{dt} + \frac{1}{LC} q = \frac{E_0 \sin \omega t}{L} \quad \text{--- (2)}$$

We have, Standard form of forced oscillation,

$$\frac{d^2 x}{dt^2} + \frac{b}{m} \frac{dx}{dt} + \frac{k}{m} x = F_p \sin \omega t \quad \text{--- (3)}$$

Comparing (2) & (3),

$$\frac{b}{m} = \frac{R}{L}$$

$$\frac{k}{m} = \frac{1}{LC}$$

$$F_p = \frac{E_0}{L}$$

Soln of (3),

$$x = A \sin(\omega''t + \delta) \quad \text{--- (4)}$$

$$\frac{dx}{dt} = v, \quad \frac{d^2x}{dt^2} = a$$

$$A = \frac{F_D}{\sqrt{m^2(\omega''^2 - \omega^2)^2 + b^2\omega''^2}} \quad \text{--- (5)}$$

Comparing
we get,

$$q_0 = \frac{E_0/L}{\sqrt{L^2(\omega''^2 - \omega^2)^2 + R^2\omega''^2}} \quad \text{--- (6)}$$

Also, from (2) & (3),

$$q = q_0 \sin(\omega''t + \delta).$$

$$\Rightarrow \frac{dq}{dt} = q_0 \omega'' \cos(\omega''t + \delta).$$

$$\Rightarrow I = I_0 \cos(\omega''t + \delta) \quad \text{--- (7)}$$

When,

$$I_0 = q_0 \omega'' \quad \text{--- (8)}$$

$$\Rightarrow I_0 = \frac{E_0/L}{\sqrt{L^2(\omega''^2 - \omega^2)^2 + R^2\omega''^2}}$$

$$\Rightarrow I_0 = \frac{E_0/L \omega''}{\omega'' \sqrt{L^2(\omega''^2 - \omega^2)^2 + R^2}}$$

$$\Rightarrow I_0 = \frac{E_0/L}{\sqrt{(\omega''L - \omega^2L)^2 + R^2}}$$

$$\Rightarrow I_0 = \frac{E_0/L}{\sqrt{\left(\omega''L - \frac{1}{\omega''C}\right)^2 + R^2}} \quad \rightarrow \text{where, } \omega^2 = \frac{1}{LC}$$

$$\Rightarrow I_0 = \frac{E_0/L}{\sqrt{(X_L - X_C)^2 + R^2}} \quad \begin{array}{l} X_L = \omega''L \\ X_C = \frac{1}{\omega''C} \end{array}$$

$$\Rightarrow I_0 = \frac{E_0/L}{\sqrt{Z^2}} \quad \text{---}$$

$$\Rightarrow \boxed{I_0 = \frac{E_m}{Z}} \quad E_m = \frac{E_0}{L}$$

Where,

$$Z = \sqrt{(X_L - X_C)^2 + R^2}$$

= Impedance,

X_L = Inductive reactance

X_C = Capacitive reactance.

When,

$X_L = X_C \rightarrow$ Condition of resonance.

$$\omega''L = \frac{1}{\omega''C}$$

$$\omega''^2 = \frac{1}{LC}$$

$$\omega''^2 = \omega^2$$

$$\therefore \boxed{\omega'' = \omega}$$

Quality Factor

$$Q = \frac{\text{voltage across (C or L)}}{\text{voltage across R}}$$

$$\Rightarrow Q = \frac{V_L}{V_R}$$

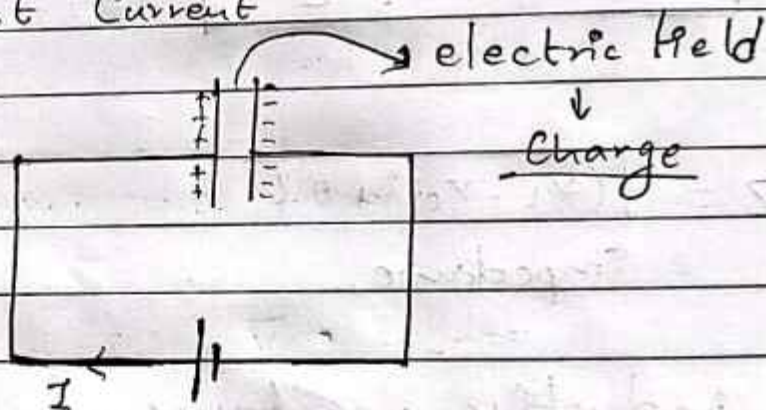
$$\Rightarrow Q = \frac{X_L}{R}$$

$$\text{or, } Q = \frac{X_C}{R}$$

$$\Rightarrow Q = \frac{\omega'' L}{R}$$

$$Q = \frac{1}{\omega'' C R}$$

Displacement Current



Total current = current due to flow of charge
+ current due to change in field.

$$\Rightarrow I_{\text{total}} = I + I_d$$

$$I = \frac{q}{L}$$

$$I_d = \frac{1}{\epsilon_0} \times \text{change enclosed (q)}$$

$$\Rightarrow EA = \frac{q}{\epsilon_0}$$

$$\Rightarrow E = \frac{q}{A\epsilon_0}$$

$$\Rightarrow \frac{d\epsilon}{dt} = \frac{1}{A\epsilon_0} \frac{d\varphi}{dt}$$

$$\Rightarrow \frac{d\epsilon}{dt} = \frac{1}{A\epsilon_0} \cdot I_d$$

$$\therefore I_d = \epsilon_0 A \cdot \frac{d\epsilon}{dt} \quad \text{or} \quad I_d = \epsilon_0 \cdot \frac{d\varphi}{dt}$$

* Different types of integral

• $\oint \vec{A} \cdot d\vec{l}$ = line integral of vector A.

• $\int \vec{A} \cdot d\vec{a}$ = surface integral of vector A.

• $\int \vec{A} \cdot d\vec{v}$ = volume integral of vector A.

* surface integral gives flux of a vector.

$$\int \vec{B} \cdot d\vec{a} = \text{Flux}$$

$$\Rightarrow B \int da$$

$$= BA$$

$$\int \vec{B} \cdot d\vec{a} = \Phi_m$$

$\times$ - curl
 $\cdot$ - divergence.

$\nabla \rightarrow$ Nabla

curl $\rightarrow$ cross product

* Stoke's Law.

$$\oint \vec{A} \cdot d\vec{l} = \int \vec{\nabla} \times \vec{A} \cdot d\vec{a}$$

$\vec{\nabla}$ = three dimensional operator

$$\vec{\nabla} = \vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z}$$

$$\Rightarrow \oint \vec{A} \cdot d\vec{l} = \int \text{curl of } \vec{A} \cdot d\vec{a}$$

$$\text{curl } \vec{A} = \vec{\nabla} \times \vec{A}$$

* Gauss - divergence law:

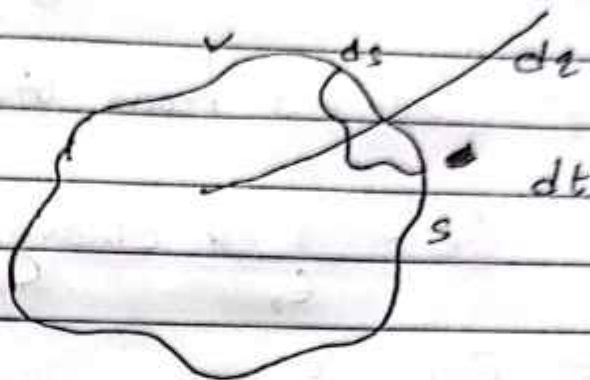
$$\int_V \vec{A} \cdot d\vec{a} = \int_V \vec{\nabla} \cdot \vec{A} \cdot dV$$

$$= \int_V \text{divergence of } \vec{A} \cdot dV$$

Where,

$$\text{divergence of } \vec{A} = \vec{\nabla} \cdot \vec{A}$$

Equation of Continuity



$$I = - \frac{dq}{dt} \quad \text{--- (1)}$$

$$\vec{j} = \frac{d\vec{r}}{ds}$$

Again,

$$j = \frac{dq}{dv}$$

$$\Rightarrow \int dq = \int j dv$$

$$\Rightarrow q = \int_V j dv \quad \text{--- (2)}$$

$$\Rightarrow \int dI = \int \vec{j} \cdot d\vec{s}$$

$$\Rightarrow I = \int_S \vec{j} \cdot d\vec{s} \quad \text{--- (3)}$$

Substituting (2) & (3) in (1),

$$\int_S \vec{j} \cdot d\vec{s} = - \frac{\partial}{\partial t} \int_V j dv$$

$$\Rightarrow \int_V (\text{div } \vec{j}) dv = - \frac{\partial}{\partial t} \int_V j dv$$

$$\Rightarrow \text{div } \vec{j} \int_V dv = - \frac{\partial j}{\partial t} \int_V dv$$

$$\Rightarrow \text{div } \vec{j} = - \frac{\partial j}{\partial t}$$

$$\Rightarrow \boxed{\nabla \cdot \vec{j} = - \frac{\partial \rho}{\partial t}}$$

It shows conservation of charge.

Maxwell's Electromagnetic Wave Equations.

1) 1st eqⁿ (Derived from Gauss law of electrostatics)

$$Q_e = \frac{1}{\epsilon_0} \times \text{charge enclosed.}$$

$$\Rightarrow EA = \frac{q}{\epsilon_0}$$

$$\Rightarrow \int_S \vec{E} \cdot d\vec{a} = \frac{q}{\epsilon_0} \quad \text{--- (1)}$$

Also,

$$q = \int_V \rho \, dv \quad \text{--- (2)}$$

From (1) & (2), we get,

$$\int_S \vec{E} \cdot d\vec{a} = \frac{1}{\epsilon_0} \int_V \rho \, dv.$$

$$\Rightarrow \int_V (\text{div } \vec{E}) \, dv = \frac{1}{\epsilon_0} \int_V \rho \, dv$$

$$\Rightarrow \text{div } \vec{E} = \frac{\rho}{\epsilon_0}$$

$$\Rightarrow \nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

Divergence $\times$ electric field

$\rightarrow$ prove

$\Phi_m = \text{magnetic flux}$

classmate



2nd eqⁿ (Derived from Gauss law of magnetism)

$$\Phi_m = BA$$

$$\text{or } \Phi_m = \int_S \vec{B} \cdot d\vec{a}$$

$$\text{but } \Phi_m = 0$$

$$\int_S \vec{B} \cdot d\vec{a} = 0$$

$$\int_V (\text{div } \vec{B}) dv = 0$$

$$\Rightarrow \text{div } \vec{B} = 0$$

$$\Rightarrow \vec{\nabla} \cdot \vec{B} = 0$$



3) 3rd eqn (Derived from Faraday's law of EM induction)

$$e = - \frac{d\phi_m}{dt} \quad \text{--- (1)}$$

e.m.f = work done to move 1 col. charge round a ckt.

$$e = \oint \vec{E} \cdot d\vec{l} \quad \text{--- (2)}$$

Work = force $\times$ disp.

$$dW = E \times dl$$

$$W = \oint E \cdot dl$$

= line integral
electric field
= e.m.f.

$$\phi_m = \int_S \vec{B} \cdot d\vec{a} \quad \text{--- (3)}$$

from (1), (2) and (3),

$$\oint \vec{E} \cdot d\vec{l} = - \frac{\partial}{\partial t} \int_S \vec{B} \cdot d\vec{a}$$

$$\Rightarrow \int_S (\text{curl } \vec{E}) \cdot d\vec{a} = - \frac{\partial}{\partial t} \int_S \vec{B} \cdot d\vec{a}$$

$$\Rightarrow \text{curl } \vec{E} = - \frac{\partial \vec{B}}{\partial t}$$

$$\Rightarrow \vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}$$

#

4th eqⁿ (Derived from Ampere's Law).

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 (I + I_d) \quad \text{--- ①}$$

$$\Rightarrow \oint \vec{B} \cdot d\vec{l} = \mu_0 I + \mu_0 I_d$$

$$\Rightarrow \oint \vec{B} \cdot d\vec{l} = \mu_0 \int_S \vec{J} \cdot d\vec{a} + \mu_0 \int_S \vec{J}_d \cdot d\vec{a}$$

$$\Rightarrow \int_S (\text{curl } \vec{B}) \cdot d\vec{a} = \mu_0 \int_S (\vec{J} + \vec{J}_d) \cdot d\vec{a}$$

$$\Rightarrow \text{curl } \vec{B} = \mu_0 (\vec{J} + \vec{J}_d)$$

$$\Rightarrow \boxed{\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{d\vec{E}}{dt}}$$

↓

In free space.

$$\begin{aligned} \frac{I_d}{A} &= \epsilon_0 \frac{dE}{dt} \\ \Rightarrow J_d &= \epsilon_0 \frac{dE}{dt} \end{aligned}$$

$$\boxed{\vec{\nabla} \times \vec{B} = \mu_0 \epsilon_0 \frac{d\vec{E}}{dt}}$$

vector triple product Rule

$$\vec{A} \times (\vec{B} \times \vec{C})$$

$$(\vec{A} \cdot \vec{C})\vec{B} - (\vec{A} \cdot \vec{B})\vec{C}$$

Speed of Electromagnetic Waves in free space.

→ From maxwell's 3rd eqⁿ,

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

Taking curl on both sides,

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = -\frac{\partial (\vec{\nabla} \times \vec{B})}{\partial t}$$

$$\text{or, } (\vec{\nabla} \cdot \vec{E})\vec{\nabla} - (\vec{\nabla} \cdot \vec{\nabla})\vec{E} = -\frac{\partial}{\partial t} \left(\mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right)$$

$$\text{or, } -\nabla^2 \vec{E} = -\mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\therefore \nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} \quad \text{--- (*)}$$

Electromagnetic wave eqⁿ in electric vector form.

Similarly,

We can write,

$$\nabla^2 \vec{B} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2} \quad \text{--- (**).}$$

Electromagnetic wave eqⁿ in magnetic vector form.

$$\left[\begin{array}{l} \hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1 \\ \hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0 \end{array} \right]$$

$$\nabla^2 = \nabla \cdot \nabla = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right)$$

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

In one dimensional form eqⁿ (*) can be written,

$$\frac{\partial^2 E}{\partial x^2} = \mu_0 \epsilon_0 \frac{\partial^2 E}{\partial t^2} \quad (***)$$

Similarly,

$$\frac{\partial^2 B}{\partial x^2} = \mu_0 \epsilon_0 \frac{\partial^2 B}{\partial t^2} \quad (****)$$

We know,

general wave eqⁿ is,

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{u^2} \cdot \frac{\partial^2 y}{\partial t^2} \quad (*****)$$

Comparing (3) and (5) or (4) and (5).

$$\mu_0 \cdot \epsilon_0 = \frac{1}{u^2}$$

$$\text{or } u = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ m/s}$$

$$\boxed{u = c}$$

In dielectric medium,

$$u = \frac{1}{\sqrt{\epsilon \mu}}$$

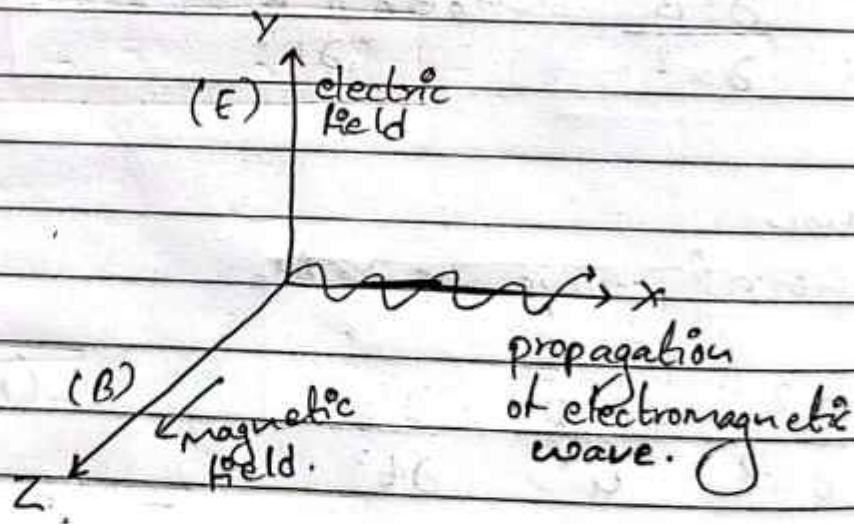
$$\therefore u = \frac{1}{\sqrt{\epsilon_r \epsilon_0 \mu_r \mu_0}}$$

$$\therefore u = \frac{1}{\sqrt{\mu_0 \epsilon_0} \cdot \sqrt{\mu_r \epsilon_r}}$$

$$\text{or, } u = \frac{c}{\sqrt{\mu_r \epsilon_r}}$$

$$\therefore \underline{u < c}$$

Relation Betⁿ Electric Field, Magnetic Field and Speed of Electromagnetic Wave.



$$\therefore E_y = E_0 \sin(\omega t - kx) \quad \text{--- (1)}$$

It is the solⁿ of electromagnetic wave in electric vector form.

i.e.

$$\frac{\partial^2 E_y}{\partial x^2} = \mu_0 \epsilon_0 \frac{\partial^2 E_y}{\partial t^2} \quad \text{--- (2)}$$

$$[\hat{i} \times \hat{j} = \hat{k}] \quad [\hat{k} \times \hat{j} = -\hat{i}]$$

P P
P P
P P
P P

$$\therefore B_z = B_0 \sin(\omega t - kx) \quad \text{--- (3)}$$

Also,

$$\frac{\partial^2 B_z}{\partial x^2} = \mu_0 \epsilon_0 \cdot \frac{\partial^2 B_z}{\partial t^2} \quad \text{--- (4)}$$

It is the electromagnetic wave eqⁿ in magnetic vector form.

We have, Maxwell's 3rd eqⁿ,

or.

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\text{or, } \left(\hat{i} \cdot \frac{\partial}{\partial x} + \hat{j} \cdot \frac{\partial}{\partial y} + \hat{k} \cdot \frac{\partial}{\partial z} \right) \times \hat{j} E_y = -\frac{\partial \hat{k} \cdot B_z}{\partial t}$$

$$\text{or, } \hat{k} \cdot \frac{\partial E_y}{\partial x} - \hat{i} \cdot \frac{\partial E_y}{\partial z} = -\hat{k} \cdot \frac{\partial B_z}{\partial t}$$

$$\therefore \frac{\partial E_y}{\partial x} = -\frac{\partial B_z}{\partial t} \quad \text{--- (5)}$$

Substituting (1) & (2) in (5),

$$\frac{\partial}{\partial x} [E_0 \sin(\omega t - kx)] = -\frac{\partial}{\partial t} [B_0 \sin(\omega t - kx)]$$

$$E_0 \cos(\omega t - kx) (-k) = -B_0 \cos(\omega t - kx) \cdot \omega$$

$$\text{or, } E_0 \cdot k = B_0 \cdot \omega$$

$$\frac{E_0}{B_0} = \frac{\omega}{k} = \frac{2\pi f}{2\pi/\lambda}$$

$$\text{or, } \frac{E_0}{B_0} = \frac{f}{\lambda}$$

$$\therefore \frac{E_0}{B_0} = c$$

$\therefore E$ & B fields are velocity selectors.

Numerical:

- Q) A ckt has $L = 5 \text{ mH}$ and $C = 2 \mu\text{F}$. How much resistance must be inserted in the ckt to reduce the resonant frequency by 5%.

Solⁿ

We have,

Resonant frequency,

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \quad \text{--- (i)}$$

If resistor R is inserted then, -

$$f' = \frac{1}{2\pi\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}} \quad \text{--- (ii)}$$

$$\boxed{f' = f_0 - 5\% \text{ of } f_0} \quad \text{--- (iii)}$$

Now, from (i) & (iii)

$$\begin{aligned} f' &= \frac{1}{2\pi\sqrt{LC}} - 5\% \text{ of } \frac{1}{2\pi\sqrt{LC}} \\ &= \frac{1}{2\pi\sqrt{5 \times 10^{-3} \times 2 \times 10^{-6}}} - \frac{5}{100} \times \frac{1}{2\pi\sqrt{5 \times 10^{-3} \times 2 \times 10^{-6}}} \\ &= 1511.96 \end{aligned}$$

Now,

Again in eqⁿ (ii),

$$1511.96 = \frac{1}{2\pi \sqrt{\frac{1}{3 \times 10^{-3} \times 2 \times 10^{-6}} - \frac{R^2}{4 \times (5 \times 10^{-3})^2}}}$$

$$1511.96 =$$

$$f' = f_0 - \frac{s}{100} f_0$$

$$\Rightarrow f' = \frac{95 f_0}{100}$$

$$\Rightarrow \frac{f'}{f_0} = \frac{95}{100} \quad \text{--- (10)}$$

$$\frac{1}{2\pi \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}} = \frac{95}{100}$$

$$\frac{1}{2\pi \sqrt{LC}}$$

$$\frac{\sqrt{LC}}{\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}} = \frac{95}{100}$$

$$\frac{LC}{\frac{1}{LC} - \frac{R^2}{4L^2}} = \frac{95^2}{100^2}$$

$$(5 \times 10^{-3}) \times (2 \times 10^{-6}) = 0.9025 \left(\frac{1}{LC} - \frac{R^2}{4L^2} \right)$$

$$\text{or } \frac{1 \times 10^{-8}}{0.9025} = \frac{1}{LC} - \frac{R^2}{4L^2}$$

$$\text{or, } \frac{R^2}{4L^2} = 10,00,00,000$$

$$\text{a, } R^2 = 10,0000000 \times 4 \times (5 \times 10^{-3})^2$$

$$\text{a, } R^2 = 10,000$$

$$\therefore R = 100 \Omega$$

#

Q. If 10 mH inductor and two capacitors of $5\mu\text{F}$ and $2\mu\text{F}$ are given, find the two resonant frequencies that can be obtained by connecting these elements in different ways.

(i) Parallel combination.

(ii) series combination.

Solⁿ

Given,

$$C_1 = 5\mu\text{F}$$

$$C_2 = 2\mu\text{F}$$

$$L = 10\text{ mH} = 10/1000 = 0.01\text{ Hz}$$

$$f_0 = ?$$

For equivalent capacitance in parallel,

$$C = C_1 + C_2$$

$$= 5 + 2$$

$$= 7\mu\text{F}$$

Resonant freq, we have,

$$f_{0\text{ parallel}} = \frac{1}{2\pi\sqrt{LC}}$$

$$= \frac{1}{2\pi\sqrt{0.01 \times 7 \times 10^{-6}}}$$

$$= 601.55\text{ Hz}$$

For equivalent capacitance in series,

$$C = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2}}$$

$$= \frac{C_1 C_2}{C_1 + C_2}$$

$$= \frac{5 \times 2}{5 + 2}$$

$$= 1.428 \mu F$$

Resonant freq,

$$f_0 \text{ series} = \frac{1}{2\pi\sqrt{LC}}$$

$$= \frac{1}{2\pi\sqrt{0.01 \times 1.428 \times 10^{-6}}}$$

$$= 1331.85 \text{ Hz}$$

What is the displacement current through a parallel plate capacitor with circular plates of radius 5 cm when the field intensity in the region betⁿ plates is varying at $10^{12} \text{ Vm}^{-1}\text{s}^{-1}$?

Solⁿ

$$I_d = ? \quad (\text{Displacement current})$$

$$r = 5 \text{ cm (radius)} = 0.05 \text{ m}$$

$$\frac{\partial E}{\partial t} = 10^{12} \text{ Vm}^{-1}\text{s}^{-1}$$

(Variable electric field)

Now,

$$(\epsilon_0 = 8.85 \times 10^{-12})$$

$$\begin{aligned} A &= \pi r^2 \\ &= \pi \times (0.05)^2 \\ &= 7.85 \times 10^{-3} \text{ m}^2 \end{aligned}$$

here,

$$I_d = \epsilon_0 \cdot A \cdot \frac{\partial E}{\partial t}$$

$$\therefore I_d = 8.85 \times 10^{-12} \times 7.85 \times 10^{-3} \times 10^{12}$$

$$\therefore I_d = 6.947 \times 10^{-2} \text{ A}$$

A light beam travelling in the x-direction is described by the electric field E_y ,
 $E_y = 300 \text{ v/m} \sin \omega \left(t - \frac{x}{c} \right)$. An electron is constrained to move along the y-direction with a speed of $2 \times 10^7 \text{ m/s}$. Find the maximum electric and magnetic force experienced by the electron.

Sol:

Given,

$$E_y = 300 \sin \omega \left(t - \frac{x}{c} \right) \quad \text{--- (1)}$$

$$c = 2 \times 10^7 \text{ m/s}$$

Comparing eqⁿ (1) with,

$$E_y = E_0 \sin(\omega t - kx)$$

$$\bullet \text{ Max}^m \text{ EF} = E_0 = 300,$$

$$\bullet \quad kx = \omega x / c$$

$$k = \omega / c$$

Now,

$$\begin{aligned} \text{Max}^m \text{ Magnetic field } (B_0) &= \frac{E_0}{c} \\ &= \frac{300}{2 \times 10^7} \\ &= 1.50 \times 10^{-5} \end{aligned}$$

$$\text{Max}^m \text{ electric force } (F_e) = q \cdot E_0$$

$$= (1.6 \times 10^{-19}) 300$$

$$= 4.8 \times 10^{-17} \text{ N}$$

$$\text{Max}^m \text{ magnetic force } (F_m) = B_0 q \cdot v$$

$$= 1.50 \times 10^{-5} \times 1.6 \times 10^{-19} \times 2 \times 10^7$$

$$= 4.8 \times 10^{-17} \text{ N}$$

Quantum Mechanics



Photon energy,
 $E = h\nu$ ——— (1)

$h \rightarrow$ planck's constant $= 6.62 \times 10^{-34} \text{ Js}$

$\nu \rightarrow$ frequency

$$E = mc^2 \text{ ——— (2)}$$

$$c = \nu \times \lambda$$

from (1) & (2),

$$mc^2 = h\nu$$

$$\Rightarrow mc^2 = h \frac{c}{\lambda}$$

$$\therefore \lambda = \frac{h}{mc}$$

$$\therefore \lambda = \frac{h}{p} \text{ for photon.}$$

For other particle,

$$\lambda = \frac{h}{mv}$$

$$\Rightarrow \lambda = \frac{h}{p}$$

classmate
Date _____
Page _____

* Velocity of Wave (phase velocity ' V_p ') -----

$$V_p = \nu \lambda \text{ ----- } (1)$$

Also,

$$\nu = \frac{mc^2}{h} \text{ ----- } (2)$$

$$\text{Also, } \lambda = \frac{h}{mv} \text{ ----- } (3)$$

$$\therefore V_p = \frac{mc^2}{h} \cdot \frac{h}{mv}$$

$$V_p = \frac{c^2}{v}$$

$$\Rightarrow \boxed{V_p = c \times \frac{c}{v}}$$

$V_p > c \rightarrow$ It is not possible.

* Group Velocity

$$V_g = \frac{d\omega}{dk} \text{ ----- } (1)$$

We have, $\omega = 2\pi\nu$

$$\Rightarrow \omega = 2\pi \cdot \frac{mc^2}{h} \text{ ----- } (2)$$

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}} \text{ ----- } (3)$$

from ② & ③,

$$W = \frac{2\pi c^2}{h} \frac{m_0}{\sqrt{1 - v^2/c^2}}$$

Diff. w.r.t. v ,

$$\frac{dW}{dv} = \frac{2\pi c^2 m_0}{h} \left(-\frac{1}{2}\right) \left(\frac{1 - v^2}{c^2}\right)^{-3/2} \left(-\frac{2v}{c^2}\right)$$

$$\Rightarrow \frac{dW}{dv} = \frac{2\pi m_0 v}{h \left(1 - \frac{v^2}{c^2}\right)^{3/2}} \text{ ----- ④}$$

Also, we have, $k = \frac{2\pi}{\lambda}$ ----- ⑤

And, $\lambda = \frac{h}{mv}$ ----- ⑥

$$\therefore k = \frac{2\pi mv}{h}$$

$$\Rightarrow k = \frac{2\pi v}{h} \cdot \frac{m_0}{\sqrt{1 - v^2/c^2}} \text{ ----- ⑦}$$

$$\therefore \frac{dk}{dv} = \frac{2\pi m_0}{h \left(1 - \frac{v^2}{c^2}\right)^{3/2}} \text{ ----- ⑧}$$

Dividing eqⁿ (4) by (8),

$$\frac{d\omega/d\nu}{dk/d\nu} = v$$
$$\Rightarrow \frac{d\omega}{dk} = v$$
$$\Rightarrow \boxed{V_g = v} \text{ --- (9)}$$

Again,

We can show the relationship betⁿ v_p & V_g .
We have,

$$v_p = v\lambda \text{ --- (10)}$$

$$\Rightarrow v = \frac{v_p}{\lambda} \text{ --- (11)}$$

Also,

$$V_g = \frac{d\omega}{dk} \text{ --- (12)}$$

Now,

$$\omega = 2\pi\nu$$

$$\Rightarrow \omega = 2\pi \frac{v_p}{\lambda} \text{ --- (13)}$$

Diff. eqⁿ (13) w.r.t. λ ,

$$\frac{d\omega}{d\lambda} = \frac{2\pi}{\lambda} \cdot \frac{dv_p}{d\lambda} - 2\pi \cdot \frac{v_p}{\lambda^2}$$
$$\Rightarrow \frac{d\omega}{d\lambda} = -\frac{2\pi}{\lambda^2} \left[v_p - \lambda \cdot \frac{dv_p}{d\lambda} \right] \text{ --- (14)}$$

Again,

$$k = \frac{2\pi}{\lambda}$$

$$\text{or } \frac{dk}{d\lambda} = -\frac{2\pi}{\lambda^2} \quad \text{--- (15)}$$

Dividing eqⁿ (14) by (15),

$$\frac{dw/d\lambda}{dk/d\lambda} = v_p - \lambda \frac{dv_p}{d\lambda}$$

$$\Rightarrow \boxed{v_g = v_p - \lambda \cdot \frac{dv_p}{d\lambda}}$$

$$\left\{ \begin{array}{l} v_g < v_p \\ v_g = v \end{array} \right\}$$

Wave length in terms of energy,

$$\lambda = \frac{h}{mv} \quad (x)$$

$$KE = \frac{1}{2} mv^2$$

$$\Rightarrow E = \frac{1}{2} mv^2$$

$$\Rightarrow v = \sqrt{\frac{2E}{m}} \quad (xx)$$

$$\therefore \lambda = \frac{h}{\sqrt{2mE}} \quad (xxx)$$

Also, $E = eV$
energy in electron volt.

Wave Function (ψ) shy

$$\psi = Ae^{-i(\omega t - kx)}$$

Wave function

It characterizes the
De-Broglie wave.

$|\psi|$ = probability density $\rightarrow$ probability of finding
the particle in somewhere.

$$\int |\psi|^2 dx = 1 \quad \rightarrow \text{there is probability}$$
$$= 0 \quad \rightarrow \text{there is no-probability.}$$

normalized
condition (1 huda).

$P \rightarrow$ position dependent.

$$\psi = Ae^{-ix}$$

normalized the function.

$$\int |\psi|^2 dx = 1$$

Also,

$$|\psi|^2 = \psi \times \psi^*$$

complex conjugate $\psi^* = Ae^{ix}$

Schrodinger's Wave Eqn^s.

1) Time dependent.

$$\psi = Ae^{-i(\omega t - kx)} \quad \text{--- (1)}$$

We have,

$$\omega = \frac{2\pi E}{h} \quad \text{--- (2)}$$

$$k = \frac{2\pi P}{h} \quad \text{--- (3)}$$

$$\therefore \psi = Ae^{-i \left[\frac{2\pi E}{h} t - \frac{2\pi P}{h} x \right]}$$

$$\Rightarrow \psi = Ae^{-\frac{2\pi i}{h} [Et - Px]}$$

$$\Rightarrow \psi = Ae^{-i/\hbar (Et - Px)} \quad \text{--- (4)}$$

$$\lambda = h/p$$

$$E = h\nu$$

$$\omega = 2\pi\nu = 2\pi \cdot E/h$$

$$\text{Also, } k = \frac{2\pi}{\lambda}$$

$$k = \frac{2\pi \cdot p}{h}$$

Where, $\hbar = \frac{h}{2\pi} \rightarrow$ This is only representation, there is no any physical meaning of $\hbar$.

(∇ is just a rep. of $\mathcal{PE}$)

If E be the total energy of particle, which has wave associated it,

Then,

$$E = K.E + P.E$$

$$\sim E = \frac{1}{2}mv^2 + V$$

$$\text{or, } E = \frac{1}{2m} p^2 + V$$

$$\text{or, } E\psi = \frac{p^2\psi}{2m} + V\psi \dots \dots \textcircled{5}$$

Now, diff. eqⁿ (4) w.r.t. 't'.

$$\frac{\partial \psi}{\partial t} = A e^{\frac{-i}{\hbar}(Et - px)} \times \left(-\frac{iE}{\hbar}\right)$$

$$\Rightarrow \frac{\partial \psi}{\partial t} = -\frac{iE}{\hbar} \psi$$

$$\Rightarrow E\psi = -\frac{\hbar}{i} \cdot \frac{\partial \psi}{\partial t}$$

$$\Rightarrow E\psi = i\hbar \frac{\partial \psi}{\partial t} \dots \dots \textcircled{6}$$

Comparing we get,

$$E = i\hbar \cdot \frac{\partial}{\partial t}$$

↓

Energy operator.

Again,

classmate
Date _____
Page _____

diff. eqⁿ (4), w.r.t. 'x'.

$$\frac{\partial \psi}{\partial x} = A e^{-\frac{i}{\hbar}(Et - px)} \cdot \left(\frac{ip}{\hbar}\right) \dots \dots \textcircled{7}$$

Again, diff. w.r.t. 'x'.

$$\frac{d^2 \psi}{dx^2} = A e^{-\frac{i}{\hbar}(Et - px)} \cdot \left(\frac{i^2 p^2}{\hbar^2}\right) \dots \dots \textcircled{8}$$

from eqⁿ (7),

$$\begin{aligned} \frac{\partial \psi}{\partial x} &= \frac{i \cdot p \cdot \psi}{\hbar} \\ \Rightarrow p \psi &= \frac{\hbar}{i} \cdot \frac{\partial \psi}{\partial x} \dots \dots \textcircled{9} \end{aligned}$$

Comparing, we get,

$$p = \frac{\hbar}{i} \cdot \frac{\partial}{\partial x}$$

↓
momentum operator.

Again, from (8),

$$\begin{aligned} \frac{\partial^2 \psi}{\partial x^2} &= \frac{i^2}{\hbar^2} \cdot p^2 \psi \\ \Rightarrow p^2 \psi &= \frac{\hbar^2}{i^2} \cdot \frac{\partial^2 \psi}{\partial x^2} \dots \dots \textcircled{10} \end{aligned}$$

→

Using eqⁿ (6) and (10) in eqⁿ (5),

$$i\hbar \cdot \frac{\partial \psi}{\partial t} = \frac{1}{2m} \hbar^2 \cdot \frac{\partial^2 \psi}{\partial x^2} + V\psi$$

$$\Rightarrow i\hbar \cdot \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \cdot \frac{\partial^2 \psi}{\partial x^2} + V\psi$$

This is the time dependent wave eqⁿ.

2) Time Independent

We have,

Wave function,

$$\psi = A e^{-i(\omega t - kx)}$$

~~is~~

$$\Rightarrow \psi = A e^{-\frac{i}{\hbar} (Et - p_x x)} \quad \dots \quad (1)$$

We also have, time dependent wave equation,

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V\psi \quad \dots \quad (2)$$

from (1),

$$\psi = A e^{-\frac{i}{\hbar} Et} \cdot e^{\frac{ip_x x}{\hbar}}$$

$$\psi = A e^{\frac{ip_x x}{\hbar}} \cdot e^{-\frac{i}{\hbar} Et}$$

$$\Rightarrow \psi = \psi_0 e^{-\frac{i}{\hbar} Et} \quad \dots \quad (3)$$

Again, diff. eqⁿ (3) w.r.t. time;

$$\frac{\partial \psi}{\partial t} = \psi_0 \left(-\frac{iE}{\hbar} \right) \cdot e^{-\frac{i}{\hbar} Et} \quad \dots \quad (4)$$

Again, diff. eqⁿ (3), w.r.t. 'x' (twice)

$$\frac{\partial^2 \psi}{\partial x^2} = \frac{\partial^2 \psi_0}{\partial x^2} \cdot e^{-\frac{i}{\hbar} Et} \quad \dots \quad (5)$$

Substituting eqⁿ (4) and (5) in (2),

$$i\hbar \cdot \psi_0 \left(\frac{-iE}{\hbar} \right) e^{-\frac{i}{\hbar} Et} = -\frac{\hbar^2}{2m} e^{-\frac{i}{\hbar} Et} \frac{\partial^2 \psi_0}{\partial x^2} + V \psi_0 e^{-\frac{i}{\hbar} Et}$$

$$\Rightarrow \psi_0 E = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi_0}{\partial x^2} + V \psi_0$$

$$\Rightarrow \frac{\hbar^2}{2m} \frac{\partial^2 \psi_0}{\partial x^2} + (E - V) \psi_0 = 0$$

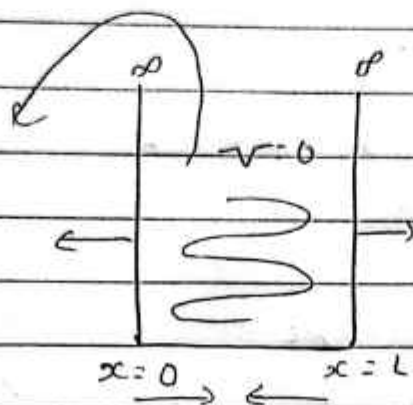
$$\Rightarrow \boxed{\frac{\partial^2 \psi_0}{\partial x^2} + \frac{2m}{\hbar^2} (E - V) \psi_0 = 0} \quad \text{--- (6)}$$

↓

time independent wave equation.

Application of Schrodinger eqⁿ:

Particle in one-dimensional potential well
(Particle in a box).



$$\left. \begin{array}{l} V=0, \quad 0 < x < L \\ V=\infty, \quad 0 \geq x \geq L \end{array} \right\} \text{--- (1)}$$

(Wave function is defined only inside the box).

$$\frac{\partial^2 \psi_0}{\partial x^2} + \frac{2m}{\hbar^2} (E - V) \psi_0 = 0 \text{ --- (2)}$$

$$\therefore \frac{\partial^2 \psi_0}{\partial x^2} + \frac{2mE}{\hbar^2} \psi_0 = 0$$

$$\Rightarrow \frac{\partial^2 \psi_0}{\partial x^2} + k^2 \psi_0 = 0 \text{ --- (3) } k^2 = \frac{2mE}{\hbar^2}$$

$$\text{Where, } k^2 = \frac{2mE}{\hbar^2} \text{ --- (4)}$$

The solⁿ of eqⁿ (3) can be written as,

$$\psi_0 = A \sin kx + B \cos kx \text{ --- (5)}$$

When, $x = 0$, $\psi_0 = 0$.

$$0 = A \sin 0 + B \cos 0.$$

$$\Rightarrow B = 0 \text{ ----- (6)}$$

$\therefore$ From (5),

$$\psi_0 = A \sin kx \text{ ----- (7)}$$

When,

$$x = L, \psi_0 = 0.$$

$$0 = A \sin kL$$

$$\text{or } \sin kL = 0, [\because A \neq 0]$$

$$\Rightarrow \sin kL = \sin n\pi, [n = 1, 2, 3, \dots]$$

$$\Rightarrow kL = n\pi$$

$$\Rightarrow k = \frac{n\pi}{L} \text{ ----- (8)}$$

from (8) & (4),

$$\frac{n^2 \pi^2}{L^2} = \frac{2mE}{\hbar^2}$$

$$\Rightarrow E = \frac{n^2 \pi^2 \hbar^2}{2mL^2} \text{ ----- (9)}$$

(Eigen values) $\rightarrow$ It shows energy is quantized.

classmate
Date _____
Page _____

Q.1) A mass of ball 10 gm has velocity 100 cm/s. Calculate the wavelength associated with it. Why does not this wave nature show up in our daily observation?
Given ($h = 6.62 \times 10^{-34}$ J-sec.)

Solⁿ

$$m = 10 \text{ gm}$$

$$v = 100 \text{ cm/s}$$

We know,

$$\lambda = \frac{h}{mv}$$

$$= \frac{6.62 \times 10^{-34}}{10 \times 100}$$

$$= \frac{6.62 \times 10^{-34}}{1000}$$

$$= 6.62 \times 10^{-32} \text{ m}$$

#

Q) Calculate the velocity and de-Broglie wavelength of a proton of energy 10^5 electron volt such that mass of proton is 1.66×10^{-27} kg charge of the proton 1.6×10^{-19} cal. and Planck's constant 6.61×10^{-34} J-sec.

Solⁿ

$$KE = \frac{1}{2} mv^2$$

$$KE = (10^5 \times 1.6 \times 10^{-19}) \text{ J}$$

$$\text{or } 10^5 = \frac{1}{2} \times 1.66 \times 10^{-27} \times (v)^2$$

$$\text{or } v^2 = 1.20 \times 10^{32} \times 1.6 \times 10^{-19}$$

$$\therefore v = 4.39 \times 10^{-6} \text{ m/s}$$



P
P
P
P
P

~~$$v = 1.09 \times 10^{16} \text{ m/s} \times 1.6 \times 10^{-19} = 1.7 \times 10^{-3} \text{ m/s}$$~~

Now,

$$\lambda = \frac{h}{mv} = \frac{6.62 \times 10^{-34}}{1.66 \times 10^{-27} \times 4.39 \times 10^{-6}}$$

~~$$\therefore \lambda = 2.84 \times 10^{-4} \text{ m}$$~~

$$\therefore \lambda = 0.0908 \text{ m}$$

Q) What is the energy of gamma photon having a wavelength of $1 \text{ \AA}$ and given planck's constant, $h = 6.62 \times 10^{-34} \text{ J-s}$.

Sol

$$\lambda = 1 \text{ \AA} = 1 \times 10^{-10} \text{ m.}$$

$$h = 6.62 \times 10^{-34} \text{ J-s.}$$

Here,

$$E = h\nu$$

$$\text{or } E = \frac{hc}{\lambda}$$

$$\text{or } E = \frac{6.62 \times 10^{-34} \times 3 \times 10^8}{1 \times 10^{-10}}$$

$$\therefore E = 1.986 \times 10^{-15} \text{ J}$$

Q.4) A fast moving neutron is found to have an associated de-Broglie Wavelength of $2 \times 10^{-12} \text{ m}$. Find the K-E, phase velocity and group velocity of de-Broglie wave.
Sol

$$\lambda = 2 \times 10^{-12} \text{ m}$$

$$E = ?$$

$$V_p = ?$$

$$V_g = ?$$

Now,

$$E = \frac{1}{2} mv^2 \quad \text{--- (1)}$$

$$\text{or } E = \frac{1}{2m} \cdot p^2$$

$$\left[v = \frac{p}{m} \right]$$

$$\text{or } E = \frac{h^2}{2m\lambda^2}$$

$$\left[p = \frac{h}{\lambda} \right]$$

$$\text{or } E = \frac{(6.62 \times 10^{-34})^2}{2 \times 1.67 \times 10^{-27} \times (2 \times 10^{-12})}$$

$$\therefore E = 1.31 \times 10^{-6} \text{ J}$$

Here,

$$E = h\nu$$

$$\text{or } \nu = \frac{E}{h}$$

$$\text{or } \nu = \frac{1.31 \times 10^{-6}}{6.62 \times 10^{-34}}$$

$$\therefore \nu = 1.97 \times 10^{27} \text{ Hz}$$

$$\therefore V_p = \frac{w}{k}$$

$$\text{or } V_p = \sqrt{\lambda}$$

$$\text{or } V_p = \frac{E}{h} \times \lambda$$

$$\text{or } V_p = \frac{1.31 \times 10^{-6}}{6.62 \times 10^{-34}} \times 2 \times 10^{-12}$$

$$\therefore V_p = 3.95 \times 10^{15} \text{ m/s}$$

Now,

$$V_g = V \text{ --- --- --- } (2)$$

Now,

$$\lambda = \frac{h}{mv}$$

$$\text{or } v = \frac{h}{m\lambda}$$

$$\text{or } v = \frac{6.62 \times 10^{-34}}{1.67 \times 10^{-27} \times 2 \times 10^{-12}}$$

$$\therefore v = 1.9 \times 10^5 \text{ m/s} = V_g$$

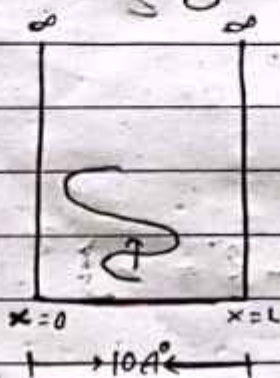
04/02

Himal

Date _____
Page _____

Q. A particle is moving one dimensional box of infinite height of width $10 \text{ \AA}$. Calculate the probability of finding the particle within an interval of $1 \text{ \AA}$ at the center of the box, when it is in least energy state.

Soln:



$$\Psi = \sqrt{\frac{2}{L}} \cdot \sin \frac{n\pi x}{L} \quad \text{--- (1)}$$

$$n = 1, \quad \Delta x = 1 \text{ \AA}$$

Now,

Probability, $|\Psi|^2 \Delta x$

$$P = \frac{2}{L} \sin^2 \frac{n\pi x}{L} \cdot \Delta x$$

$$P = \frac{2}{L} \sin^2 \frac{\pi}{L} \cdot \frac{L}{2} \cdot \Delta x \quad (n=1)$$

$$P \Rightarrow \frac{2}{L} \sin^2 \frac{\pi}{2} \cdot \Delta x$$

$$P \Rightarrow \frac{2 \cdot \Delta x}{L}$$

$$P = \frac{2}{10 \times 10^{-10}} \times 10^{-10}$$

$$\therefore P = 0.2$$

Q) The wave function of electron moving an x -axis like a wave is given by $\psi(x) = 2 \sin 2\pi x$. Calculate the probability of finding out an electron from 0.25m to 0.5m , along the length on axis.

Soln:

$$\int_{0.25}^{0.5} |\psi|^2 dx$$

$$= \int_{0.25}^{0.5} (2 \sin 2\pi x)^2 dx$$

$$= \int_{0.25}^{0.5} 4 \sin^2 2\pi x dx$$

$$= 4 \int_{0.25}^{0.5} \left[\frac{1 - \cos 2 \cdot 2\pi x}{2} \right] dx \quad \left[\int \sin ax = \frac{\cos ax}{a} \right]$$

$$= 0.2 \left[\left[x \right]_{0.25}^{0.5} - \left[\frac{-\sin 4\pi x}{4\pi} \right]_{0.25}^{0.5} \right]$$

$$= 2 \left((0.5 - 0.25) + \left(\frac{\sin 4\pi \cdot 0.5}{4\pi} - \frac{\sin 4\pi \cdot 0.25}{4\pi} \right) \right)$$

$$= 2 \times 0.25 + 0$$

$$= 0.5$$

$$\langle P_x \rangle = 0$$

Himal

Date _____
Page _____

$$\langle P_x \rangle$$

Q) Calculate the expectation value of P_x of the momentum of a particle trapped in one dimensional box.

Soln

$$\langle P_x \rangle = \int \psi \langle P_x \rangle \psi dx \quad \text{--- (1)}$$

but,

$$\psi = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L} \quad \text{--- (2)}$$

$$[\psi^* = \psi]$$

(only in this case)

Also,

$$P_x = \frac{h}{i} \frac{\partial}{\partial x} \quad \text{--- (3)}$$

Putting it in eqⁿ (1),

$$\langle P_x \rangle = \int_0^L \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L} \left(\frac{h}{i} \frac{\partial}{\partial x} \right) \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L} dx$$

$$= \frac{2}{L} \frac{h}{i} \int_0^L \sin \frac{n\pi x}{L} \frac{\partial}{\partial x} \left\{ \sin \frac{n\pi x}{L} \right\} dx$$

$$= \frac{2h}{L} \cdot \frac{n\pi}{L} \int_0^L \sin \frac{n\pi x}{L} \cdot \cos \frac{n\pi x}{L} \cdot dx$$

=

Q) Normalize the one dimensional wave function given by,

$$\psi(x) = A \sin\left(\frac{\pi x}{a}\right), \quad 0 < x < a$$

$$\psi(x) = 0, \text{ outside.}$$

$$\int_0^a |\psi|^2 dx = 1 \rightarrow \text{Normalized condition}$$

$$\Rightarrow A^2 \int_0^a \left[\frac{1 - \cos \frac{2\pi x}{a}}{2} \right] dx = 1.$$

$$\Rightarrow \frac{A^2}{2} \left[\int_0^a dx - \int_0^a \cos \frac{2\pi x}{a} dx \right] = 1.$$

$$\Rightarrow \frac{A^2}{2} [a] = 1.$$

$$A^2 = \frac{2}{a}$$

$$A = \sqrt{\frac{2}{a}}$$

$\therefore$ Normalized wavefunction,

$$\psi = \sqrt{\frac{2}{a}} \cdot \sin\left(\frac{\pi x}{a}\right).$$

Q. Calculate the permitted energy levels of an electron in the box of 1°A wide.

$$\left[E_n = \frac{n^2 h^2}{8mL^2} \right]$$

$$h = 6.62 \times 10^{-34}$$

$$m = 9.1 \times 10^{-31}$$

$$n = 1, 2, 3, \dots$$

$$L = 1^\circ\text{A} = 10^{-10} \text{ m}$$

Here,

$$\begin{aligned} n=1 \rightarrow E_n &= \frac{1^2 \times (6.62 \times 10^{-34})^2}{8 \times (9.1 \times 10^{-31}) \times (10^{-10})^2} \\ &= 6.01 \times 10^{-18} \end{aligned}$$

$$\begin{aligned} n=2 \rightarrow E_n &= 2 \times \\ &= 2.40 \times 10^{-17} \end{aligned}$$

$$\begin{aligned} n=3 \rightarrow E_n &= 3 \times \\ &= 5.41 \times 10^{-17} \end{aligned}$$

* Thermodynamics:

Thermodynamic process. $\left\{ \begin{array}{l} 1^{st} \text{ law} \\ 2^{nd} \text{ law} \end{array} \right\} \rightarrow$ Heat ~~energy~~, engine, heat pump, refrigerator.

Heat transfer $\rightarrow$ black body
 $\rightarrow$ Rate of energy flow
 $\rightarrow$ emissive power emissivity.

$$dQ = dU + dW$$

$\downarrow$
 - first law of thermodynamics.

Also, $dW = F \times dx$

$$dW = P \cdot A \cdot dx$$

$$\boxed{dW = P \cdot dv}$$

$$\boxed{dQ = dU + p dv}$$

* Isochoric process. ($V = \text{const}$)

$$dQ = dU + p dv$$

$$p dv = 0$$

$$\boxed{\therefore dQ = dU}$$

* Iso baric (pressure constant)

$$\boxed{dQ = dv + p \, dv}$$

$\frac{dQ}{dT}$ = heat capacity

$$\frac{C_p}{C_v} = \gamma$$

$$C_p > C_v$$

$$S.H.C = \frac{\partial Q}{m \cdot dT}$$

$$\rightarrow C_p dT = dv + p \cdot dv$$

* Iso thermal process.

$$dv = 0$$

$$\Rightarrow \boxed{dQ = p \cdot dv}$$

$$\Rightarrow \int dQ = \int_{v_1}^{v_2} p \cdot dv$$

$$\Rightarrow Q = \int_{v_1}^{v_2} \frac{c}{v} \cdot dv$$

$$Q = c \log_e \left(\frac{v_2}{v_1} \right)$$

$$pV = \text{const.}$$

$$P_1 V_1 = \text{const.}$$

$$P_2 V_2 = \text{const.}$$

$$\therefore P_1 V_1 = P_2 V_2 = c.$$

$$Q = P_1 V_1 \log_e \left(\frac{v_2}{v_1} \right)$$

or,

$$Q = P_1 V_1 \log_e \left(\frac{P_1}{P_2} \right)$$

work done in isothermal process.

At Adiabatic

$$dQ = du + p dv$$

but,

$$dQ = 0.$$

$$du + p \cdot dv = 0 \quad \text{--- (1)}$$

but,

$$du = C_v dT \quad \text{--- (2)}$$

Also,

$$PV^\gamma = \text{const.}$$

$$\boxed{PV = RT}$$

$$P = \frac{RT}{V} \quad \text{--- (3)}$$

$\therefore$ from (1), (2), & (3)

$$C_v \cdot dT + \frac{RT}{V} \cdot dv = 0.$$

$$C_v \cdot \frac{dT}{T} + \frac{R}{V} \cdot dv = 0$$

$$C_v \log_e T + R \cdot \log_e V = \text{const.}$$

$$\log_e T + \frac{R}{C_v} \log_e V = C$$

$$\log_e \frac{PV}{R} + \frac{R}{C_v} \log_e V = C$$

$$\log_e \frac{PV}{R} + (r-1) \log_e V = C$$

$$\Rightarrow \log_e \left[\frac{PV}{R} \cdot v^{v-1} \right] = \text{const.}$$

$$\Rightarrow \log_e \left(\frac{PV}{R} \right) = \text{const.}$$

$$\frac{PV}{R} = e^{\text{const.}}$$

$$\boxed{\Rightarrow PV = \text{const.}}$$

Work

$$dw = P dv$$

$$w = \int_{v_1}^{v_2} P dv$$

$$w = \int_{v_1}^{v_2} \frac{C}{v^v} dv$$

~~Q1~~

The wave function of a particle is given by,

$$\psi = ce^{-x^2/2\alpha^2}, \quad -\infty < x < \infty$$

where c and α are constant.

Calculate the probability of finding the particle in the region, $0 < x < \infty$

$$\text{Ans} \Rightarrow P = c^2 \frac{1}{2} \sqrt{\frac{\pi}{2\alpha^2}}$$

- # Black Body
- # Stefan's Boltzmann Law
- # Emissive Power and emissivity
- # Fourier's law of heat conduction

Imp. for numerical

Emissive Power,

$$P = \sigma A (T^4 - T_0^4)$$

↳ for perfect black body.

Not perfect black body,

$$P = \sigma A e (T^4 - T_0^4)$$

e = radiation energy of a body at certain temp
radiation energy of a black body at same temp.



$$\frac{dQ}{dT} \propto (T_2 - T_1)$$

$$\frac{dQ}{dT} \propto A$$

$$\frac{dQ}{dt} \propto \frac{1}{d}$$

$$\frac{dQ}{dt} \propto \frac{(T_2 - T_1)}{d}$$

~~$$\frac{dQ}{dt} \propto \frac{1}{d}$$~~

$$\frac{dQ}{dt} = \frac{kA(T_2 - T_1)}{d}$$